CHEMISTRY FORM FOUR
QUESTIONS AND ANSWERS
CHAPTER ONE
HYDEROCARBONS

1) What are hydrocarbons?

- **Hydrocarbons** are compounds that contain carbon and hydrogen only.

2) Where hydrocarbons obtained from?

- **Hydrocarbons** are mainly obtained from Natural gases, Coal and petroleum

3) What are commonly used hydrocarbons?

- **Hydrocarbons** are commonly used as fuels, candle wax, oil gasoline, raw materials in the synthesize of different consumer products including fabrics, soaps, dyes, cosmetics and drugs.

Edit by:Tisca

4) List three hydrocarbon compounds?

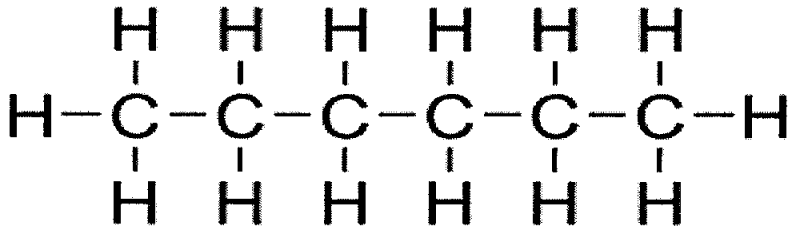
- **Methane =CH_4 ,C_3H_6=propene** and **C_5H_8= pentyne** are all hydrocarbon compound because they composed **of carbon** and **hydrogen** only.

5) On the bases of bond. What are types of hydrocarbons?

- On the basis of bond, hydrocarbons are classified as **saturated** and **unsaturated** hydrocarbons

6) What are saturated hydrocarbons?

- **Saturated hydrocarbons** are hydrocarbons that contain only C-C single bond. Example Hexane



7) What are also called saturated hydrocarbons?

- Saturated hydrocarbons are also **called alkanes.**

8) Define unsaturated hydrocarbons?

- **Unsaturated hydrocarbons** are hydrocarbons that contain C-C multiple bonds. Alkenes and alkynes are unsaturated hydrocarbons

9) On the basis of structure. What are hydrocarbons be classified?

- On the basis of structure hydrocarbons are classified as **Cyclic** and **acyclic hydrocarbons**.

Edit by:Tisca

10) What are cyclic hydrocarbons?

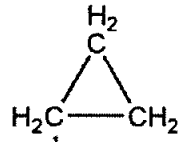
- **Cyclic hydrocarbons** are closed or ring structure of hydrocarbons.

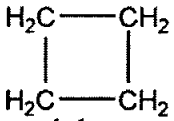
11) What are cyclic hydrocarbons sub-divided?

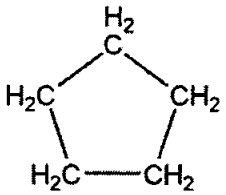
- **Cyclic hydrocarbons** are sub-divided as Alicyclic and aromatic hydrocarbons

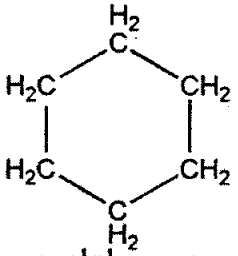
12) List parts of alicyclic hydrocarbons

a) **Cyclo-alkane**: are closed structure of alkanes

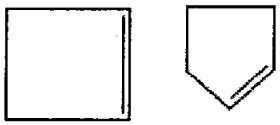

cyclopropane


cyclobutane

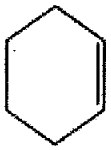

cyclopentane

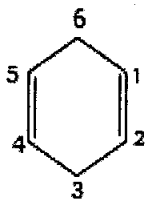

cyclohexane

b) **Cyclo-alkene**: are closed structure of alkenes


Cyclobutene

Cyclopentene

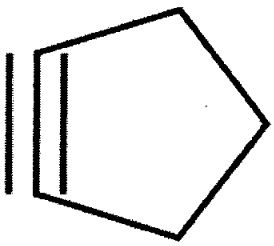

Cyclohexene

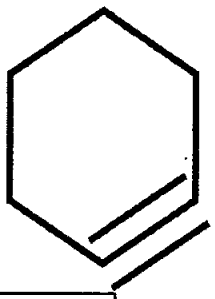


1, 4-Cyclohexadiene

Edit by:Tisca

c) **Cyclo-alkynes**: are closed structure of alkynes


Cyclopentyne

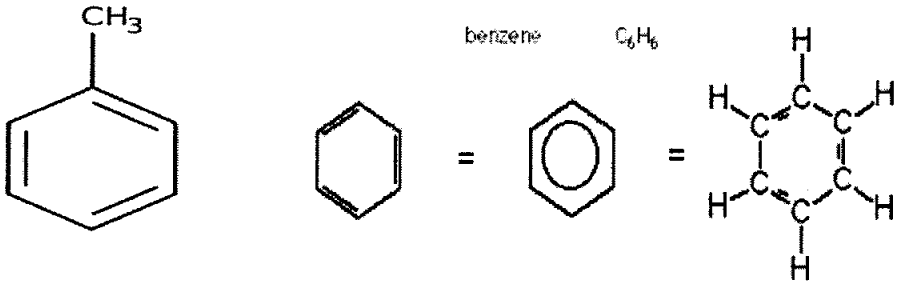

Cyclohexyne

13) What are aromatic hydrocarbons be classified?

- **Aromatic hydrocarbons** are classified as Benzoid and Non-benzoid aromatic hydrocarbons

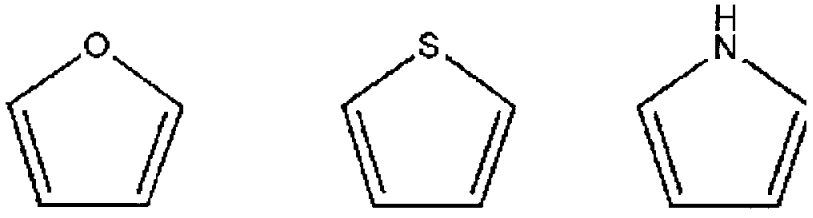
14) What are benzoid aromatic hydrocarbons?

- **Benzoid** are aromatic hydrocarbons that contain benzene ring. E.g.: Toluene



15) Define non-benzoid aromatic hydrocarbons

- **Non-benzoid** are aromatic hydrocarbons that do not contain benzene ring.



furan thiophene pyrrole

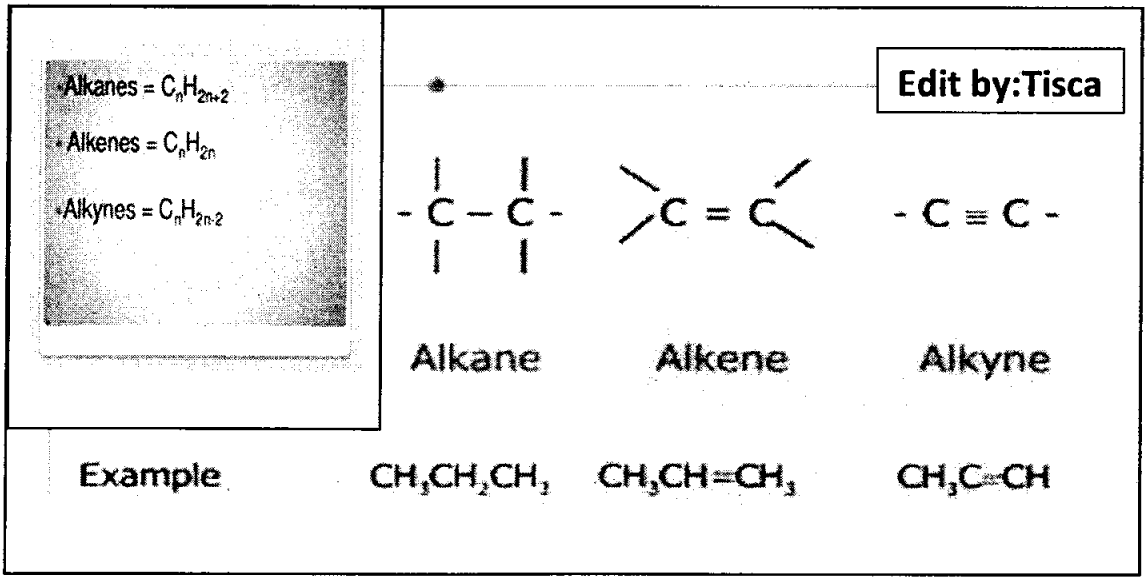
Edit by:Tisca

16) What are acyclic hydrocarbons?

- **Acyclic hydrocarbons** are open chain structure of hydrocarbons.

17) **What are acyclic hydrocarbons also called?**

- **Acyclic hydrocarbons** also called aliphatic hydrocarbons



18) **On the basis of source and properties. What are hydrocarbons be classified?**

- On the basis of their sources and properties, hydrocarbons are classified **into aliphatic** and **aromatic hydrocarbons**.

19) **What is the difference between aliphatic and aromatic hydrocarbons?**

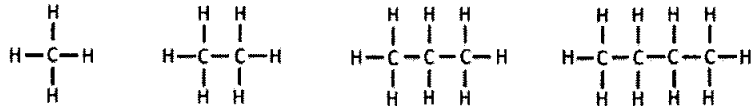
- **Aliphatic hydrocarbons** are hydrocarbons derived by the chemical degradation of fats and oils.
- **Aromatic hydrocarbons** are hydrocarbons obtained by chemical degradation of certain pleasant ant-smelling plant extracts.

20) Define alkanes

- **Alkanes** are saturated hydrocarbons containing Carbon-carbon single bond.

Alkanes C_nH_{2n+2}

methane	ethane	propane	butane
CH_4	C_2H_6	C_3H_8	C_4H_{10}



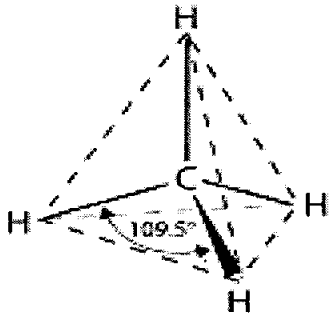
21) What are alkanes also called?

- Alkanes are also called **saturated hydrocarbons.**

22) Why alkanes are called saturated?

- Because they **saturated with hydrogen's.**

23) What is the geometric structure of alkane?



Edit by:Tisca

- The geometric structure of alkane is **tetrahedron with bond angle of 109.5°**

24) Define Sp3

- Sp^3 is the combination **one S and three P-orbitals**

25) What are alkanes be classified?

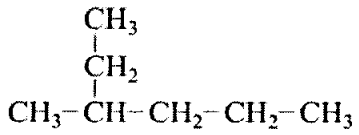
- Alkanes are classified as **straight chain alkanes** and **branched chain alkanes**

26) Define straight chain alkanes

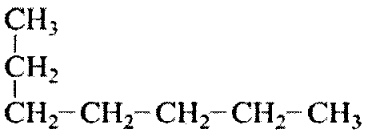
- **Straight chain alkanes** are alkanes in which each carbon atom is bonded to not more than two carbon atoms. Straight chain alkanes are also known **as normal alkanes. E.g:**
  - **CH_3-CH_2-CH_2-CH_2-CH_2-CH_3= Hexane**
  - **CH_3-CH_2-CH_2-CH_2.CH_2-CH_2-CH_3 = Heptane**

27) Define branched chain alkanes

- **Branched chain alkanes** are alkanes in which at least on carbon atom is bonded to more than two other carbon atoms.



branched chain



unbranched chain

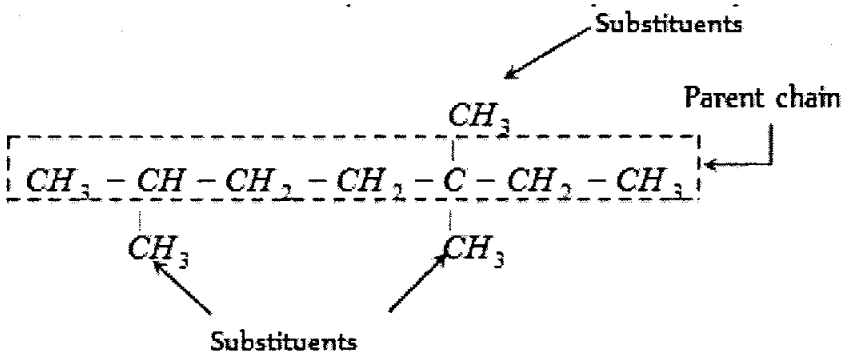
28) Structurally branched chain alkanes consist of two parts. What are they?

- Structurally branched chain alkanes consist of two parts namely: **Parent chain** and **side chain**

Edit by:Tisca

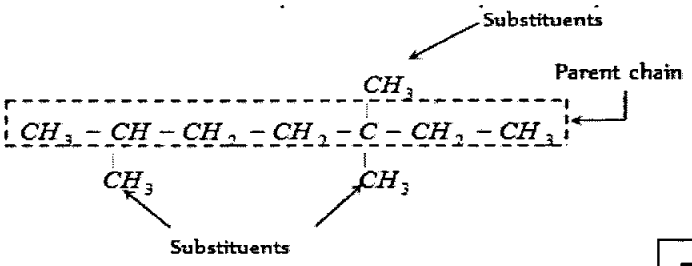
29) What is parent chain?

- **Parent chain** is the longest straight carbon-carbon chain of carbon atoms.



30) What is a side chain?

- **Side chains** are short chains attached to the parent chain. Side chains are also called **substituents or branch.**



Edit by:Tisca

31) What are alkyl groups?

- **Alkyl groups** are alkanes that have one hydrogen less. **Example**:
  Methane (CH4) =Alkanes and Methyl -CH_3 = alkyl group ,
  Ethane(CH3CH3) are alkanes and **Ethyl-CH2-CH3 is** Alkyl group.

32) What is nomenclature?

- **Nomenclature** is the naming of hydrocarbons.

33) How many ways are named in alkanes?

Alkanes are named in two different systems namely:

- **Trivial system(Common system)** and
- **IUPAC system** (International Union Pure Applied Chemistry)

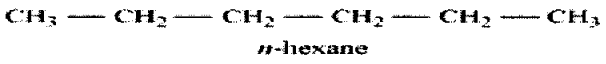
34) In common system three terms are used. What are they?

- In common system three terms are used namely:
- **Normal (n) , Iso (i)** and **Neo- (N)**

35) What is the meaning of Normal (n) , Iso (i) and Neo- (N)?

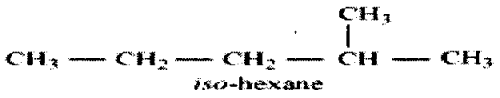
- **Normal** is used to name straight chain alkanes. Look at the first(1) image
- **Iso**-is a system used to name alkanes in which one methyl group is attached to the second carbon of the parent chain. Look at the Second (2) images.

1)



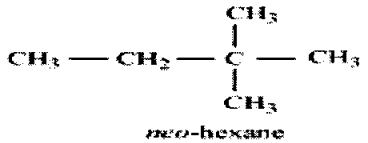
n-hexane

2)



iso-hexane

3)



neo-hexane

Edit by:Tisca

- **Neo-** is system used to name alkanes in which two methyl groups attached to the second carbon. Eg. Look at the last (3) image above

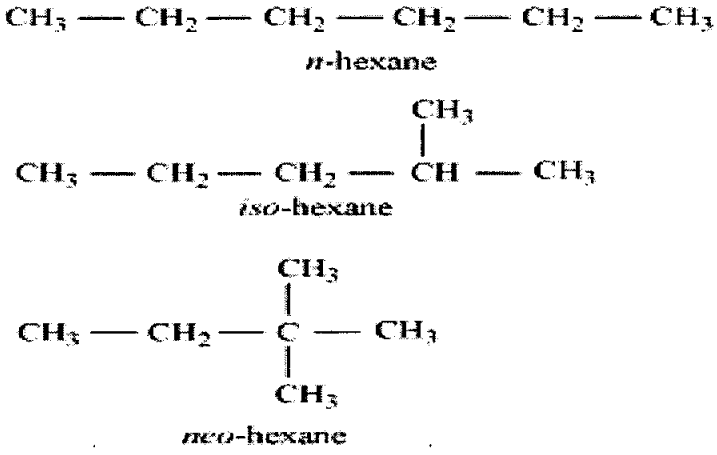
36) What is isomerism?

- **Isomerism** is the phenomenon **of** the existence **of isomers.**

37) Define Isomers?

- **Isomers** are molecules with the same molecular formula but different in structural formula.

Example: Hexane = C_6H_{14}



Edit by:Tisca

38) What is the formula used to calculate isomers?

- The number of possible isomers is calculated by $2^{n-4}+1$, $n \geq 4$.

  This formula is only applicable up to 7 carbon atoms

39) List the number of possible isomers of the first ten alkanes

Name	Molecular Formula (C_nH_{2n+2})	Condensed Structural Formula	Number of Possible Isomers
methane	CH_4	CH_4	–
ethane	C_2H_6	CH_3CH_3	–
propane	C_3H_8	$CH_3CH_2CH_3$	–
butane	C_4H_{10}	$CH_3CH_2CH_2CH_3$	2
pentane	C_5H_{12}	$CH_3CH_2CH_2CH_2CH_3$	3
hexane	C_6H_{14}	$CH_3CH_2CH_2CH_2CH_2CH_3$	5
heptane	C_7H_{16}	$CH_3CH_2CH_2CH_2CH_2CH_2CH_3$	9
octane	C_8H_{18}	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_3$	18
nonane	C_9H_{20}	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$	35
decane	$C_{10}H_{22}$	$CH_3CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_2CH_3$	75

- These formula are used above 7 carbon atoms Octane = $2^{n-4}+2$
  Nonane $2^{n-4}+3$ and Decane $2^{n-4}+11$

Edit by:Tisca

40) Write physical properties of alkanes

- Alkanes are non-polar compounds, therefore soluble in in non-polar solvents and not soluble in polar solvents.
- The first four members of straight chain alkanes (C_1-----C_4) are gases, the next thirteen are liquids (C_5------C_{17}). Those containing eighteen carbon atoms and above are wax solids.

- The boiling point of branched chain alkanes is lower than that of straight chain alkanes because the surface area of alkanes decreases with branching.
- Alkanes are lighter than water.

41) What are chemical properties of alkanes?

- **Alkanes are** generally inert towards acids, bases oxidizing and reducing agents. However alkanes are quite inert towards common reagents such as acids, alkalis and oxidizing agents.

42) What is reagent?

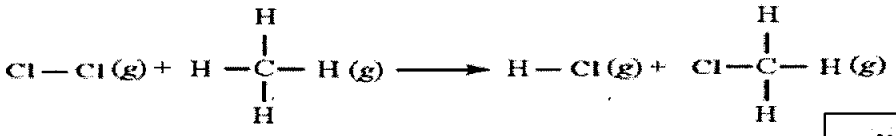
- A **reagent** is a substance that is used in **chemical** analysis and to synthesize other products.

43) List reactions that undergo alkanes

- Alkanes undergo **substitution reaction** and **combustion reaction**

44) What is substitution reaction?

- **Substitution reaction** is a reaction in which one atom is displaced by another atom. E.g. $Cl_2 + CH_4 \longrightarrow CH_3\text{-}Cl + HCl$



Edit by:Tisca

45) What is halogenation?

- Halogenations: is the addition of halogen such as Chlorine, Bromine Iodine and Fluorine

46) What is combustion reaction?

- **Combustion** reaction is the scientific word for burning. In a **combustion reaction**, a substance reacts with sufficient oxygen to form **carbon dioxide** and **water**

47) What is the general formula to balance combustion reaction?

- The general combustion reaction for alkanes can be generalized as

$$CnH2n + 2 + \frac{3n+1}{2} O2 \rightarrow nCO2 + (n+1)H2O$$

48) Practical Question: Balance the equation below by using the above formula for complete combustion reaction

[redacted] Balance the reaction for the complete combustion of hexane (C_6H_{14}) in air.

$$C_6H_{14} + O_2 \longrightarrow CO_2 + H_2O$$

Edit by:Tisca [redacted]

49) What is the product of incomplete combustion of alkane?

- **Alkanes undergo** incomplete combustion with insufficient amount of oxygen to form carbon monoxide and water.

$$CH_{4\,(g)} + O_{2\,(g)} \longrightarrow CO_{(g)} + H_2O_{(l)}$$

50) **How alkanes are prepared from?**

- **From alkyl halides (R-X)** and **Decarboxylation**
- Formation of alkanes from alky halides are two methods

a) **Reduction reaction**: in this process all **alkyl halides** except alkyl fluoride are reduced **to alkanes** in the presence of zinc and hydrochloric acid as **catalyst.**

- $CH_3Cl + H_2 \longrightarrow CH_4 + HCl$
- $C_3H_7Cl + H_2 \longrightarrow C_3H_8 + HCl$

b) **Wurtz reaction**: in this process **alky halides** react with **metallic sodium** in dry ether **to form symmetrical alkanes** with double the number carbon atoms present in the alkyl halide.

(iii) Wurtz reaction

$$CH_3 - Br + 2Na + Br - CH_3$$

Bromomethane
(methyl bromide)

$$\xrightarrow{\text{Dry ether}} CH_3 - CH_3 + 2\,NaBr$$

Ethane

Edit by:Tisca

51) What is decarboxylation?

- **Decarboxylation reaction** is a chemical **reaction** that eliminates a carboxyl group and liberates carbon dioxide (CO_2).

By Decarboxylation of Carboxylic Acid

When the sodium salt of carboxylic acid is heated strongly with sodalime (NaOH + CaO), a molecule of carbon dioxide is split of as carbonate and an alkane is formed. Notice that alkane produced contains one carbon less then the original carboxylic acid.

$$\underset{\text{Sodium salt of Acid}}{R\text{-}COONa} + NaOH \xrightarrow{\Delta} \underset{\text{Alkane}}{R\text{-}H} \quad Na_2CO_3$$

$$\underset{\text{Sodium Acetate}}{CH_3\text{-}COONa} + NaOH \xrightarrow{\Delta} \underset{\text{Methane}}{CH_4} \quad Na_2CO_3$$

Edit by:Tisca

52) What are uses of alkanes?

- **Alkanes** are used as fuel, they produce large amount of heat in combustion.
- **Liquid alkanes** are used as **solvents.**
- Higher alkanes that have **17 to 20** carbon atoms are **viscous liquid** and uses as **lubricants.**
- Solid alkanes with **21 to 30** carbon atoms are the constituents of paraffin wax used in making candles or base in cosmetics.

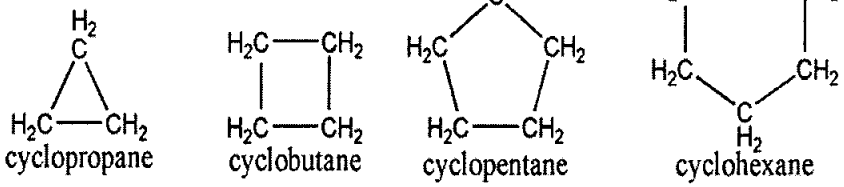
- Carbon black obtained by incomplete combustion of alkane is used in the manufacture of printing ink, black pigments, and shoe polish.

53) Why alkanes are called paraffins?

- Because they do not undergo many reactions, therefore are called paraffins. From Latin **parum affins** meaning **little affinity**.

54) What is cycloalkane?

- Cycloalkane is cyclic hydrocarbons with only single carbon-carbon bind.



55) What is the general formula of cycloalkane?

- The general formula of cycloalkane is **CnH2n**.

Edit by:Tisca

56) What are alkenes?

- **Alkanes** are **unsaturated hydrocarbons** containing at least one carbon-carbon double bond between carbon atoms

57) What is the general formula of alkenes?

- Alkenes have the general formula of CnH2n
  - **CH_2=CH-CH_2-CH_3 =1-butene and**
  - **CH2=CH-CH_2-CH_2-CH_2-CH_3 =1-Hexene**

58) What are classified into alkenes?

- Alkenes are classified as **symmetric** and **asymmetric alkenes**

59) What is the difference between symmetric and asymmetric alkenes?

- **Symmetric alkenes** are alkenes in which **two same groups** are being attached to the **carbon-carbon double bond**. E.g.:
  - CH_3 -**CH=CH**-CH_3 =2-butene
  - CH_3-CH_2-CH_2 -**CH=CH**-CH_2.CH_2-CH_3 = 4-Octene

60) What are asymmetric alkenes?

- **Asymmetric alkenes** are alkenes in which **two different groups** are being attached to the **carbon-carbon double bond**. E.g.:
  - CH_3 -**CH=CH**-CH2-CH2-CH3 =2-Hexene
  - CH3-CH2 -**CH=CH**-CH_2-CH_2.CH_2-CH_3 = 3-Octene

Edit by:Tisca

62) How alkenes are named?

- Alkenes are named in two different systems which are **common system** and **IUPAC system.**
- In common system we use appropriate prefix and we add suffix ---yline- **Ethylene, propylene , butylene, pentylene .....**

63) What is the difference between alpha and beta butylene?

- **Alpha butylene** the double bond is at carbon one.

  E.g.: CH_2=CH-CH_2-CH_2-CH_3 = **α-butylene**

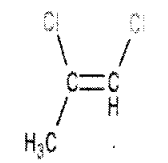
- **Beta butylene** the double bond is at carbon two.

  E.g.: CH_3 -CH_3-CH=CH-CH_3 =**β-butylene**

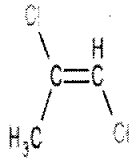
64) List isomerism in alkenes

Edit by:Tisca

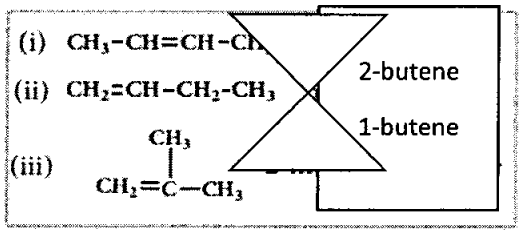
- Alkenes exhibit
  - **Chain isomerism** -is due to the difference in the arrangement of carbon atoms. (Look at **(i) and(iii) below) are chain isomerism.**
  - **Position isomerism**-is due to the difference in the location of double bond. (Look at **(i) and(ii) below) are position isomerism.**
  - **Geometrical isomerism**- is formed due to the arrangement of atoms or groups attached to the C-C double bond.



cis-1,2-dichloropropene



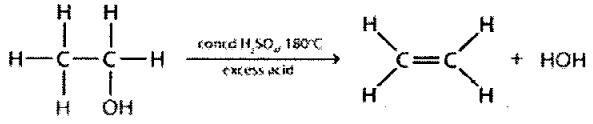
trans-1,2-dichloropropene



- **Cis= means** two similar groups attached on the same side of the double bond.
- **Trans= means** two similar groups attached on the opposite side of the double bond.

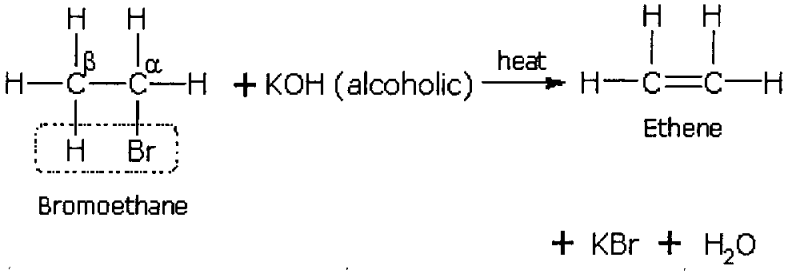
65) How alkenes are prepared?

1) **Dehydration of alcohol**: In this process alcohol is heated to produce alkenes and water.



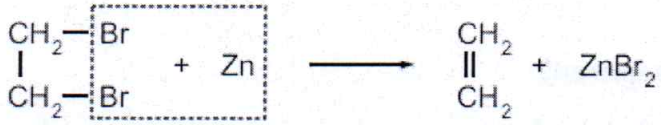
Ethanol Ethylene

2) **Dehydrohalogenation:** in this process alkyl halide is heated in alcoholic solution to form alkenes.



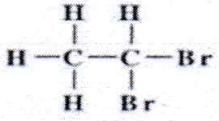
Bromoethane Ethene

3) **Dehalogenation of vicinal dihalide:** in this process vicinal dihalide is heated to produce alkenes.

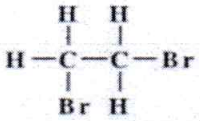


66) What is vicinal dihalide?

- **Vicinal dihalide** is dihalide that the two halogen atoms are bonded to the adjacent carbon atoms.



geminal dihalide



vicinal dihalide

67) What is germinal dihalide?

- **Germinal dihalide** is a dihalide that the two halogen atoms are bonded to the same carbon atoms.

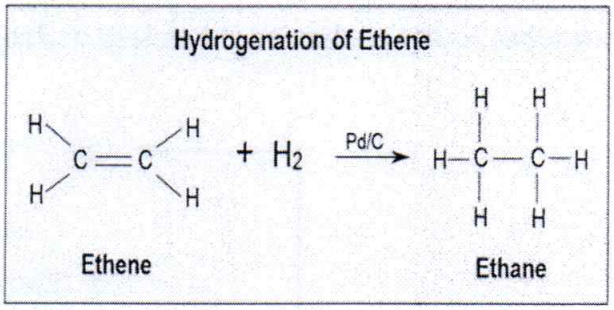
Edit by:Tisca

68) List physical properties of alkenes

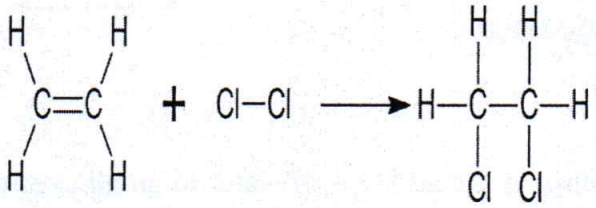
- The first three alkenes are gases, next few members are liquids and higher members are solids.
- Ethylene is colorless gas with sweet odours while others are colorless.
- They are soluble in organic solvents like benzene but insoluble in water.
- Boiling points of alkene increases with the increasing in molecular mass.
- Alkenes with branches have lower boiling points.

69) List chemical properties of alkenes

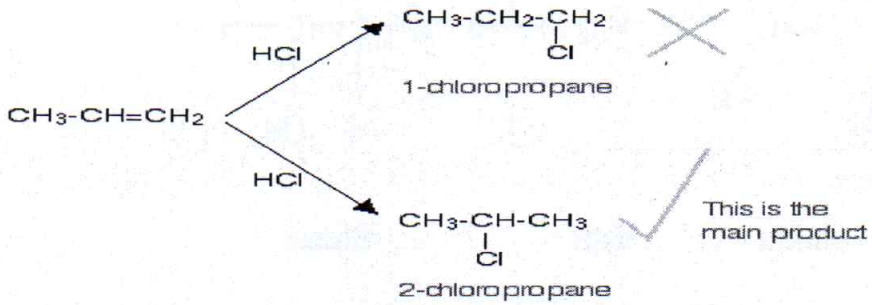
a) Hydrogenations: is the addition of hydrogen to an alkene to form alkane.



b) Halogenation: is the addition of halogens to an alkenes such as chlorine, bromine and iodine to form dihaloalkanes.



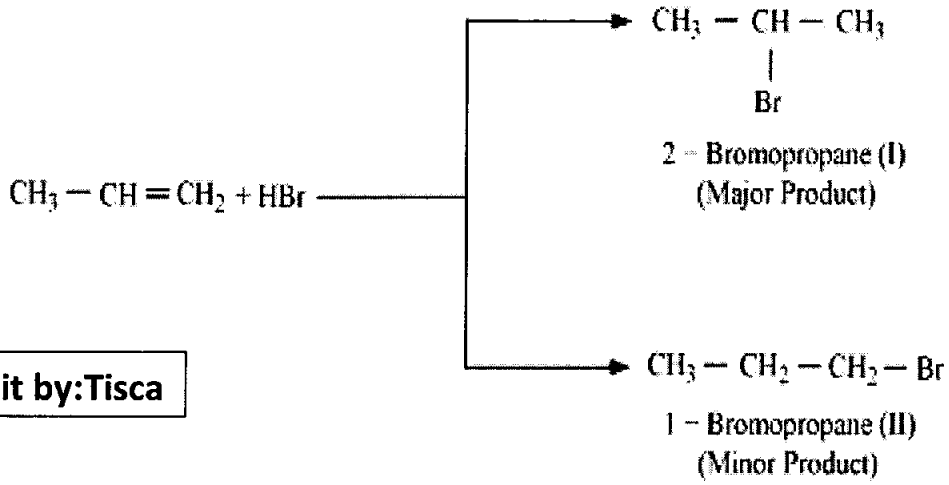
c) Hydrohalogenation: is the addition of hydrogen halides to alkenes.



Edit by:Tisca

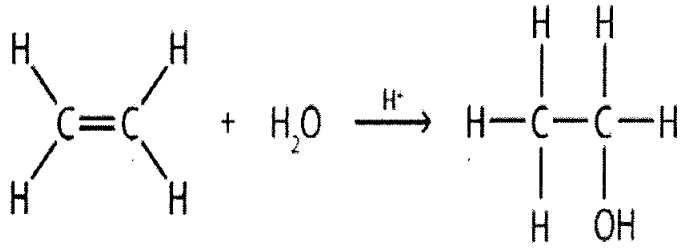
70) What does Markovinkov's rule states?

- **Markovinkov's rule** states that when **alkenes reacts with alkyl halide** The hydrogen atoms attaches **to the carbon that has more hydrogen atoms** while **halogen attaches to the carbon that has less hydrogen** atoms.



Edit by:Tisca

d) Hydration: is the addition of water to an alkenes to produce alcohol.



Ethylene Water Ethanol

e) **Combustion:** is the reaction of alkenes with sufficient and insufficient of oxygen to produce **carbon dioxide** and **carbon monoxide** respectively.

Combustion

- Complete combustion

$$C_3H_6 + O_2 \rightarrow CO_2 + H_2O$$

- Incomplete combustion

$$C_3H_6 + O_2 \rightarrow C + CO + CO_2 + H_2O$$

Edit by:Tisca

71) What is the general balanced equation of alkene?

- The general balanced equation of an alkenes is

$$CnH2n + \frac{3n}{2}O2 \rightarrow nCO2 + n\ H20$$

***Paractical* Question: Write the balanced chemical equation of**

- **C_2H_4 and C_3H_6 and C_4H_8**

72) What are uses of alkenes?

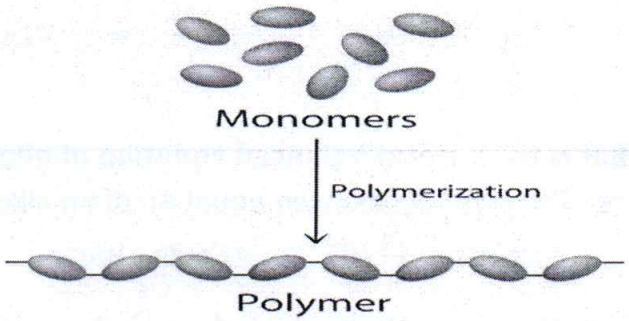
- Ethylene is used to manufacture of polyethylene
- In combustion with oxygen, ethylene is used oxyethylene flame which used for cutting and welding metals.
- Ethene is also used as a ripening agent for foods
- Alkenes are used to prepare alkanes and alcohols

73) What is polymerization

- **Polymerization:** is union of two or more monomers to form a a large molecule.

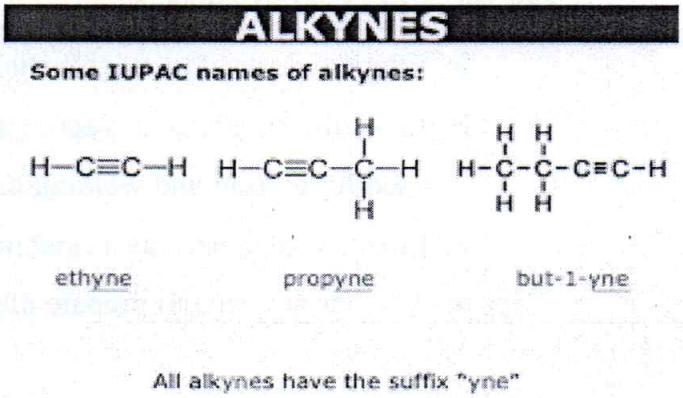
74) What is monomer?

- A monomer is the simplest unit of a polymer required to form a very large molecule



75) What are alkynes?

- **Alkynes** are unsaturated hydrocarbons that contain at least one carbon carbon-carbon triple bond.



Edit by:Tisca

76) What alkynes also called?

- Alkynes also called acetylene because the common name of the first member of alkynes is known as acetylene.

77) What is the general formula of alkynes?

- The general formula of alkynes is CnH2n-2 $n \geq 2$

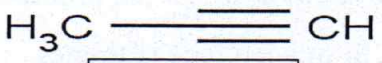
78) How alkynes are named?

- Alkynes are named in two different systems namely **Common system** and **IUPAC system**

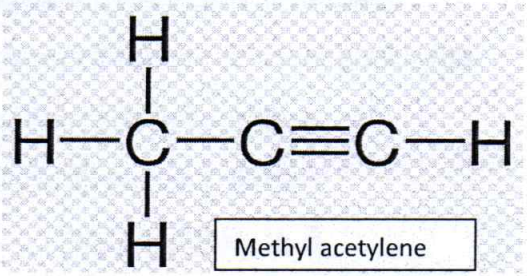
79) How common system is named in alkynes?

- In common system carbon-carbon triple bond is named as **acetylene** therefore the alkynes are named **alkyl acetylene**.

E.g:



Acetylene



Methyl acetylene

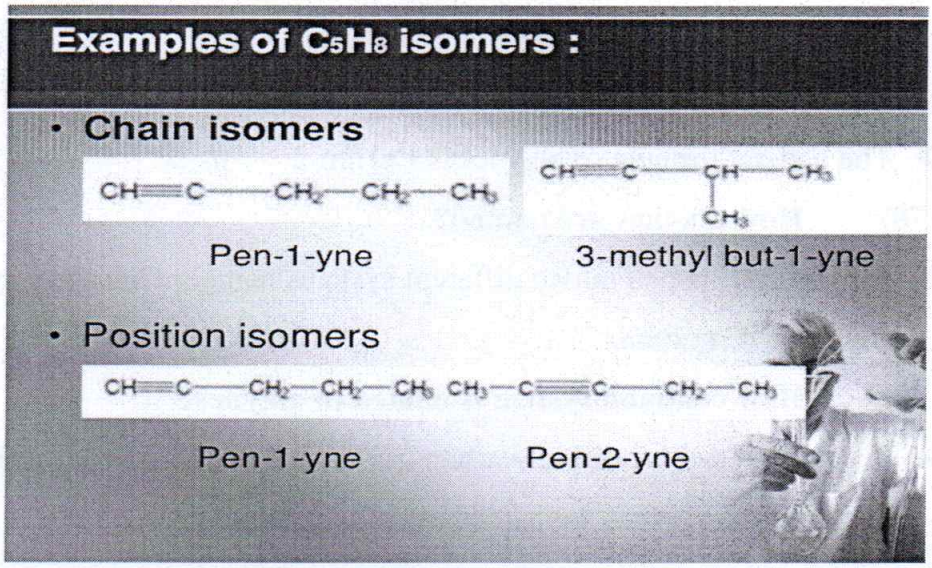
80) List Isomerism in alkynes

- Alkynes exhibit a) Chain isomerism and b) Position Isomerism

- **Chain isomerism** is due to the change in the structure of carbon atoms in the alkynes.

Edit by:Tisca

- **Position isomerism** is due to the change the location of the triple bond.



80) List general methods of preparation of alkynes

➢ **Action of water with calcium carbide**: in this process calcium carbide reacts with water to form acetylene (Ethyne).

E.g: $CaC_2 + H_20 \longrightarrow Ca(OH)_2 + C_2H_2$

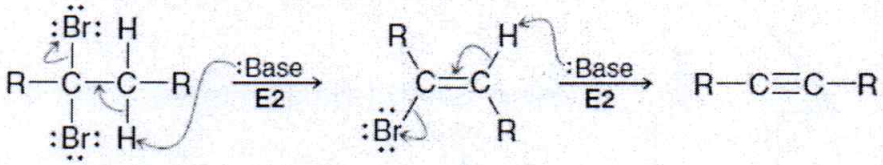
➢ **Dehydrohalogenation of vicinal and germinal dihalide**

- The loss of a hydrogen atom and a halogen atom from adjacent alkanes carbon atoms leads to the formation of an alkenes.

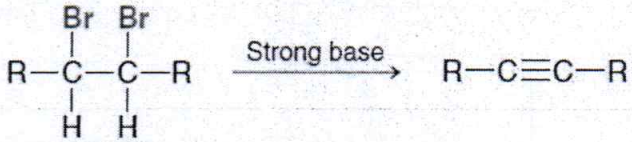
Edit by:Tisca

Preparation of Alkynes

- Such eliminations usually occur via an E2 mechanism
- **Geminal** dihalides can be used



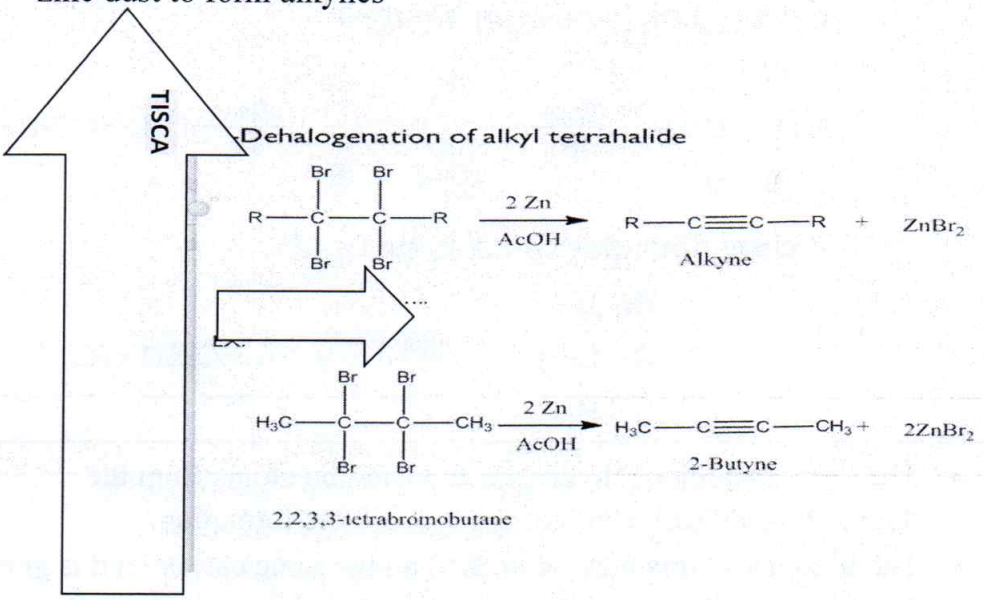
- **Vicinal** dihalides can also be used



- The loss of additional hydrogen and halogen atoms from the double-bonded carbon atoms leads to alkyne formation.
- The **halogen atoms** may be located on the same carbon that **is germinal**.
- The **halogen atoms** may be located on the adjacent carbons that are **vicinal.**

Edit by:Tisca

- **Dehalogenation of tetra halide**: In this process tetra halides are heated with zinc dust to form alkynes



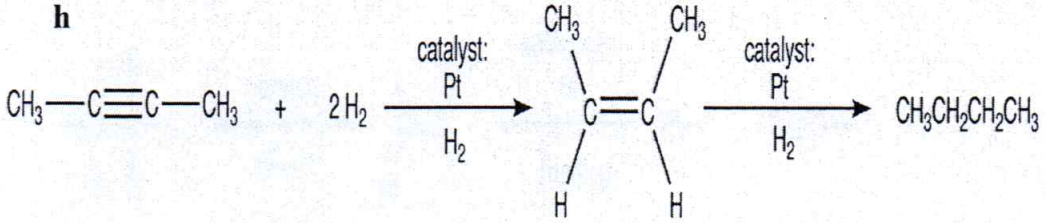
81) List uses of alkynes

- The first three alkynes are gases, the next eight members are liquids and higher members are solids.
- All alkynes except Ethyne are odorless
- Alkynes have slightly higher boiling points than corresponding alkanes and alkenes.
- The melting and boiling points increases with increase in molecular mass or number of number of carbon atoms.
- Alkynes are less dense in water and they are insoluble in water.

82) List chemical properties of alkynes

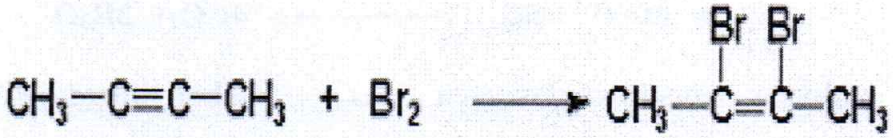
a) Hydrogenation: is the addition of hydrogen to a alkynes.

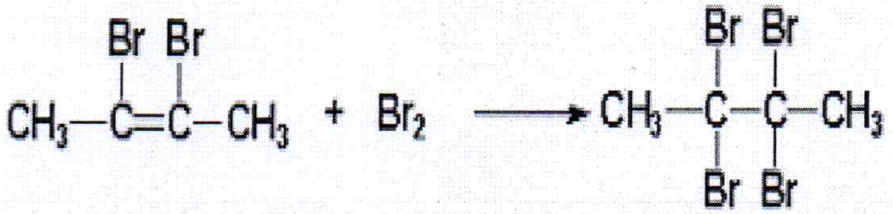
h



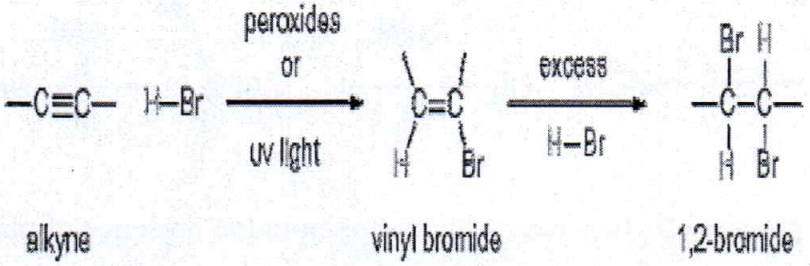
✓ In this process alkynes react with hydrogen in the presence of platinum (Pt) and Palladium (Pd) as catalyst to form alkenes. The alkenes again react with hydrogen the double bond breaks due to the addition reaction with hydrogen and alkane is formed.

b) **Halogenation:** is the addition of halogen such as chlorine or bromine dihaloalkanes. **E.g**





c) **Hydrohalogenation:** is the addition of hydrogen halide. Alkyl halides react with alkynes to form **monohaloalkenes** and then **Dihaloalkanes.**



d) **Combustion reaction:** is the reaction of alkynes with oxygen to produce a sooty, yellow flame with large amount of heat.

Alkynes + sufficient oxygen ⟶ Carbon dioxide + water

- $2C_2H_2 + 5O_2 \longrightarrow 4CO_2 + 2H_2O$

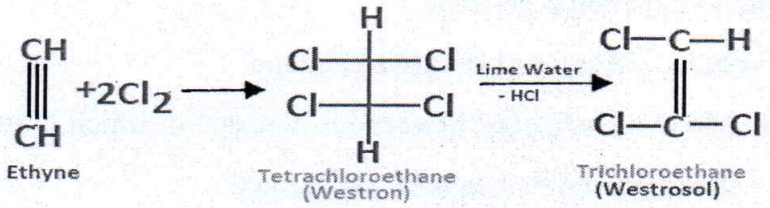
Alkynes + insufficient oxygen ⟶ Carbon dioxide +Carbon + water

- $2C_2H_2 + 3O_2 \longrightarrow 2CO_2 + 2C + 2H_2O$

83) List uses of alkynes

- Acetylene is used in Oxy-acetylene flame which used as for cutting and welding of metals.
- Acetylene is used in manufacture of westron

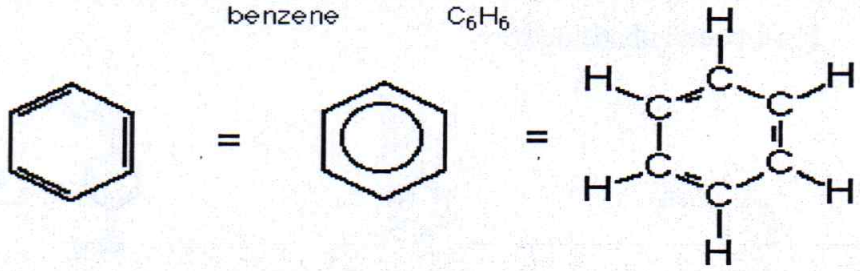
Edit by:Tisca



- Alkynes are used formation of aldehyde, ketones and alkanes.
- Alkynes are also used for manufacturing of plastics and synthetic rubber.

84) Who discovered Benzene structure?

- **Benzene structure** was discovered by **Michael faraday in 1825** but the name was given by **Fiedrich August kekule in 1865**.



- **Kekule was proved** that the benzene ring consist of alternating single and double carbon to carbon bonds.

85) What are classified benzene?

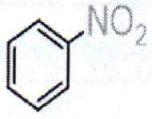
Benzene are classified as

- ✓ Mono-substituted benzene
- ✓ Di-substituted benzene
- ✓ Poly-substituted benzene

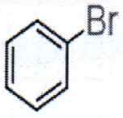
Edit by:Tisca

86) What is Mono-substituted benzene

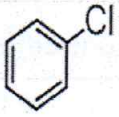
- **Mono-substituted benzene** is benzene in which only one of the hydrogen atoms has been substituted.



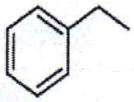
nitrobenzene



bromobenzene



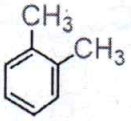
chlorobenzene



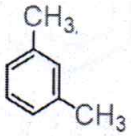
ethylbenzene

87) What is Di-substituted benzene

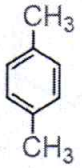
- **Di-substituted benzene** is benzene in which two hydrogen atoms have been substituted.



ortho-**Xylene**
(1,2-dimethylbenzene)



meta-**Xylene**
(1,3-dimethylbenzene)

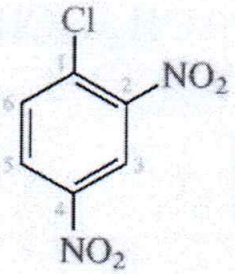


para-**Xylene**
(1,4-dimethylbenzene)

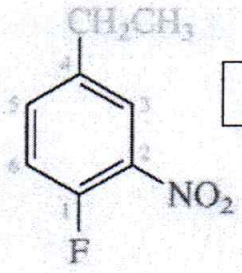
Edit by:Tisca

88) What is Poly-substituted benzene

- **Poly -substituted benzene** is benzene in which more than two hydrogen atoms have been substituted.



1-Chloro-2,4-dinitrobenzene

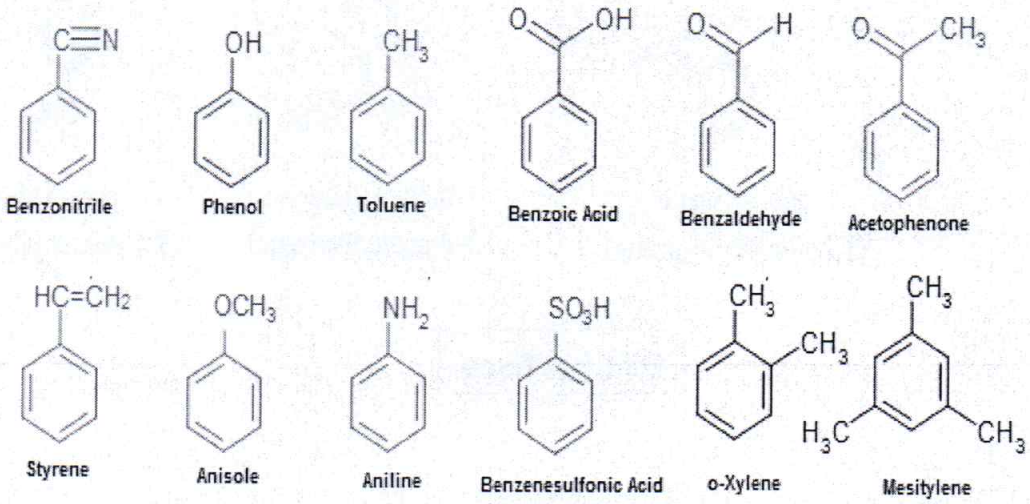


4-Ethyl-1-fluoro-2-nitrobenzene

Edit by:Tisca

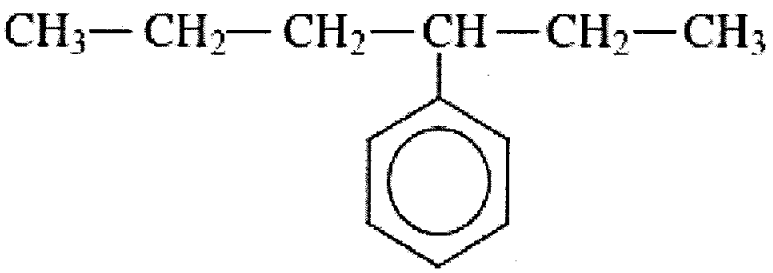
89) How common names are named in benzene?

- Mono-substituted benzene have common names that can be learned only by memorizing



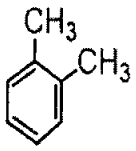
90) What is called when the benzene becomes branch?

- when the alky group larger than ethyl. The benzene becomes branch and is called phenyl group.

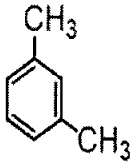


3-phenylHexane

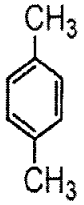
91) Give The Common and the IUPAC name of the compounds below



ortho-**Xylene**
(1,2-dimethylbenzene)



meta-**Xylene**
(1,3-dimethylbenzene)



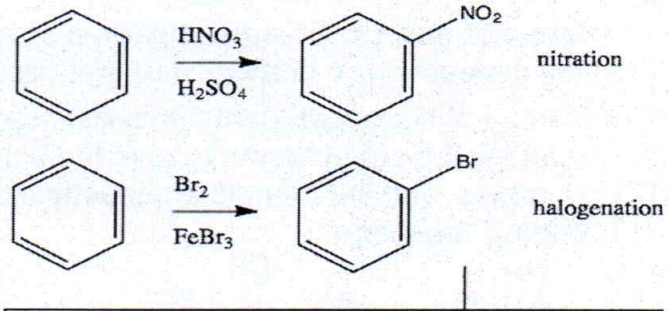
para-**Xylene**
(1,4-dimethylbenzene)

Edit by:Tisca

92) List Chemical properties of benzene

Benzene resisits addition reaction but readily undergoes

1. **Combustion reaction:** This process benzene reacts with oxygen to form carbon dioxide and water. Hence benzene it is very flammable.
   - $2C_6H_6 + 15O_2 \longrightarrow 12CO_2 + 6H_2O$
2. **Substitution reaction:** This process can be Halogenation and Nitration.
   - ✓ **Halogenation:** In this process halogen is added to the benzene with the presence of the catalysts such as ferric halide or aluminum halide
   - ✓ **Nitration:** in this process benzene is treated with a mixture of nitric acid and sulfuric acid.



1. **Nitrobenzene + H_2O**
2. **Bromobenzene+ HBr**

Edit by:Tisca

93) List Uses of benzene

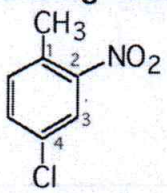
- Benzene can be converted to styrene which is used in the manufacture of synthetic fibers like nylon.
- Benzene is an important source of chemicals which are used as pesticides and detergents
- Derivatives of benzene such as toluene is used in making explosives
- Derivatives of benzene such as aniline are used as dyes and drugs.

94) How is named when one group of benzene imparts, special name?

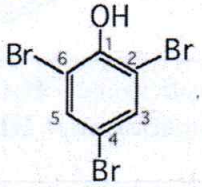
- Nomenclature — Edit by:Tisca

For three or more substituents:

- **If one of the substituents imparts a special name, name the molecule as a derivative of that parent.**
- **If none of the substituents imparts a special name, number the substituents to give the smallest set of numbers, and list them in alphabetical order before the ending "benzene".**



4-Chloro-2-nitrotoluene



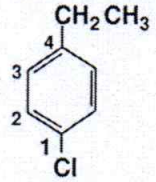
2,4,6-Tribromophenol

2-Bromo-1-ethyl-4-nitrobenzene

95) How is named benzene when neither group imparts, special name?

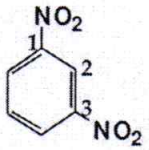
Edit by:Tisca

- where neither group imparts a special name, locate the groups and list them in alphabetical order



1-Chloro-4-ethylbenzene
(p-Chloroethylbenzene)

1-Bromo-2-nitrobenzene
(o-Bromonitrobenzene)



1,3-Dinitrobenzene
(m-Dinitrobenzene)

- ✓ Reading is the road of achievement.
- ✓ Every new day is another chance to change your life
- ✓ Try to read what you can per day
- ✓ All the best

CHEMISTRY FORM FOUR
QUESTIONS AND ANSWERS
CHAPTER TWO
ALOCOHOLS, PHENOLS AND ETHERS

1) Define functional group

- **Functional group** is an atom or group of atoms with in molecule that gives certain characteristics to the organic compounds.

2) What is the functional group of alcohol?

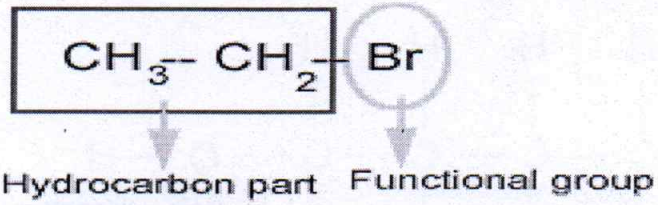
- **Alcohol** has hydroxyl group **(-OH)** as functional group **R-OH**

3) What does organic compound composed of?

Organic compounds composed of two parts

- Hydrocarbon part which is **chemically inert**
- Active part which is called **functional group**

Edit by:Tisca



4) State the function of the functional group

- **The functional group** determines the nomenclature, physical and chemical properties of the organic molecule.

5) Define alcohols

- **Alcohols** are organic compounds that contain hydroxyl group **(-OH)** as functional group.
- **Alcohol is** derivative of alkanes in which one or more hydrogen atom is replaced by hydroxyl group.

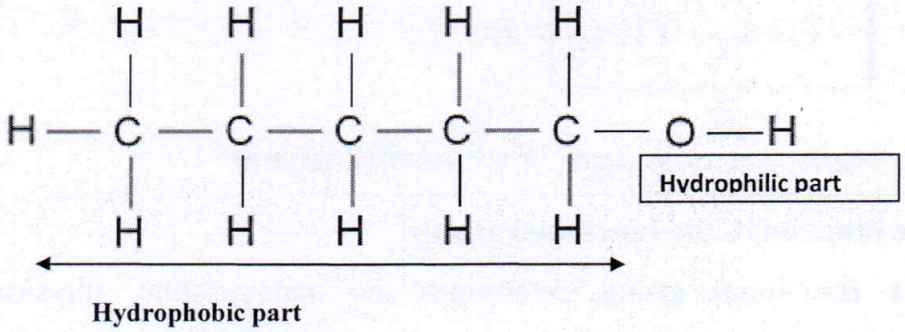
CH_3OH	CH_3CH_2OH	$CH_3CH_2CH_2OH$	$CH_3CH_2CH_2CH_2OH$
methanol	ethanol	propan-1-ol	butan-1-ol

6) State parts that alcohol consist of

Edit by:Tisca

Alcohol consist of two parts namely

- Hydrocarbon part (Alkyl group) part which is **hydrophobic. Hydrophobic** literally means "the fear of water"
- Alcohol part (Hydroxyl group) which is **hydrophilic. Hydrophilic** literally means "the love of water"

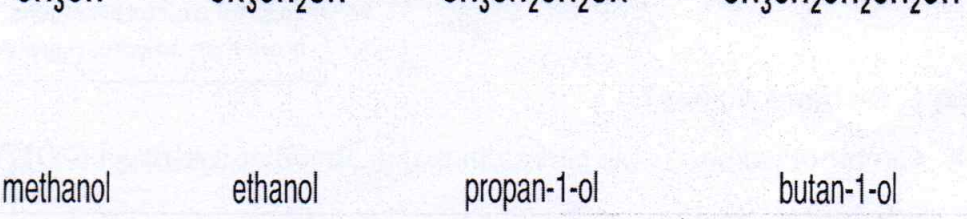


7) On the basis of hydroxyl group in the alcohol. List categories of alcohol

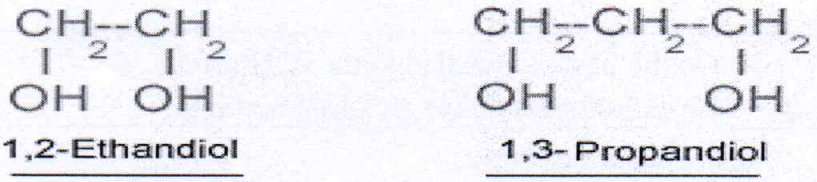
According to the hydroxyl group attached to the chain. Alcohols are classified into four categories namely.

- **Monohydric alcohol**: are monohyroxy alcohols that contain one hydroxyl group.

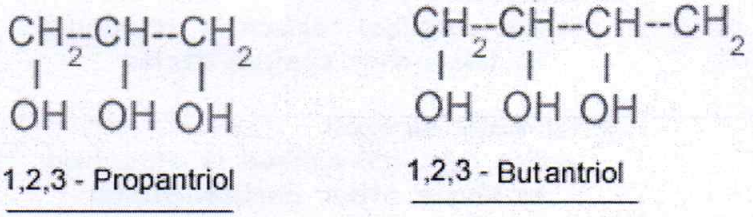
Edit by:Tisca



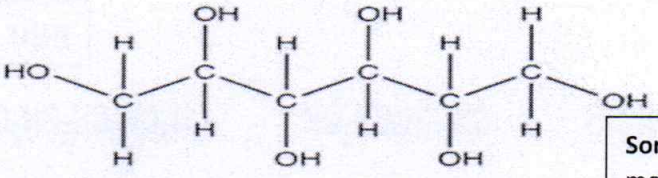
- **Dihydric alcohol**: are dihydroxy alcohols that contain two hydroxyl groups. **Dihydric alcohols** also called diols.



- **Tihydric alcohol:** are trihydroxy alcohols that contain three hydroxyl group.



- **Polyhydric alcohol**: are polyhydroxy alcohols that contain mare than three hydroxyl groups.



Edit by:Tisca

Sorbitol structure contains more than 3-hydroxyl group

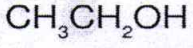
8) What is carbinol carbon?

- **Carbinol carbon** is the carbon in the alcohol that **hydroxyl (–OH)** group is being attached.

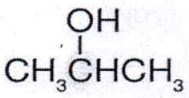
9) On the basis of Carbinol carbon. What are Monohydric alcohol be classified?

- On the basis C-bonded to the -OH group.
  Monohydric alcohol are classified into:

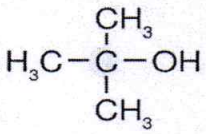
- **Primary alcohol:**
  - the carbinol carbon is attached to one other carbon atom



- **Secondary alcohol:**
  - the carbinol carbon is attached to two other carbon atoms



- **Tertiary alcohol:**
  - the carbinol carbon is attached to three other carbon atoms



10) How alcohols are named?

- **Alcohols are named by both IUPAC and Common system.**
- **In common or trivial system** the alcohol are named by combining the name of the alkyl group with the word alcohol. **E.g. Alky alcohol**

Compound	Common name	IUPAC name
$H_3C—OH$	Methyl alcohol	Methanol
$H_3C—CH_2—OH$	Ethyl alcohol	Ethanol
$H_3C—CH_2—CH_2—OH$	*n*-propyl alcohol	Propan-1-ol
$H_3C—\underset{\displaystyle OH}{\underset{\vert}{CH}}—CH_3$	isopropyl alcohol	Propan-2-ol
$H_3C—CH_2—CH_2—CH_2—OH$	*n*-butyl alcohol	Butan-1-ol
$H_3C—CH_2—\underset{\displaystyle OH}{\underset{\vert}{CH}}—CH_3$	sec-butylalcohol	Butan-2-ol
$H_3C—\underset{\displaystyle CH_3}{\underset{\vert}{CH}}—CH_2—OH$	isobutyl alcohol	2-methyl propan-1-ol
$H_3C—\overset{\displaystyle CH_3}{\overset{\vert}{\underset{\displaystyle CH_3}{\underset{\vert}{C}}}}—OH$	tert-butyl alcohol	2-methyl propan-2-ol

11) List isomerism in alcohols

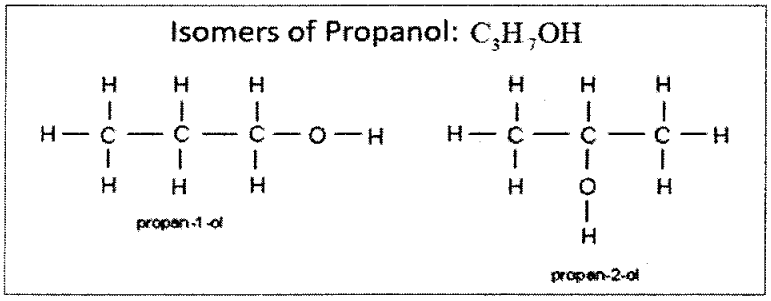
Alcohol exhibit three isomerism

- **Chain isomerism :** is due to the arrangement of carbon atoms in the molecule.

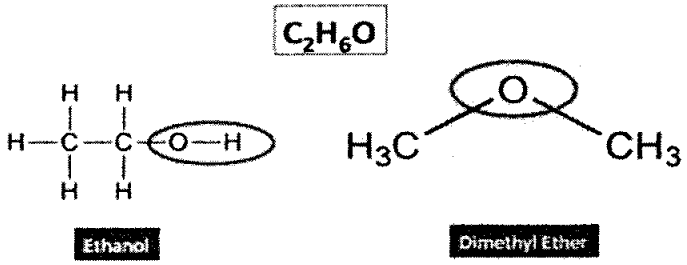
$CH_3CH_2CH_2CH_2OH$
1-Butanol

$CH_3-\underset{\displaystyle CH_3}{\underset{\vert}{CH}}-CH_2-OH$
2-methyl-1 propanol

- **Position isomerism**: is due to the position of the **-OH** group.

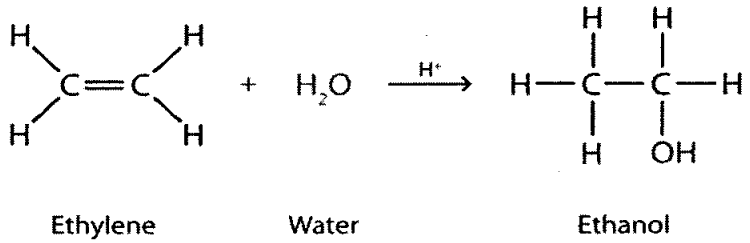


- **Functional isomerism:** in this process molecules have **same molecular** formula but **differ in functional group.**



12) How alcohols are prepared?

- Alcohols are prepared by hydration of alkenes. (**Addition of water to alkenes**)

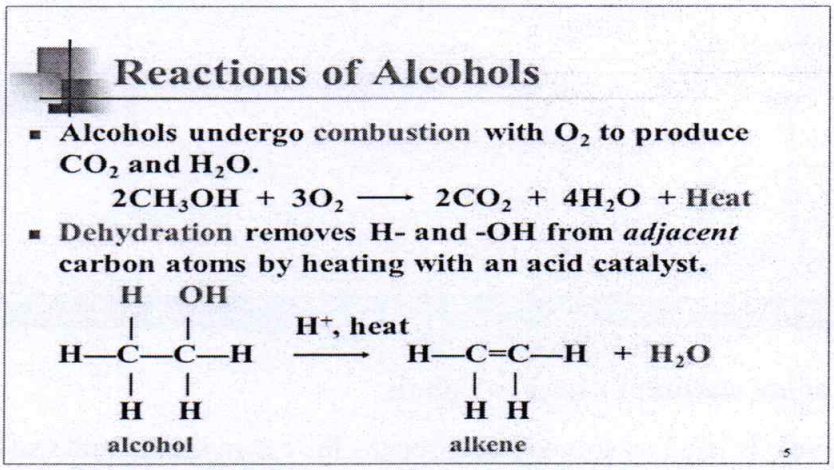


13) Write physical properties of alcohols

- The physical properties of alcohols are responsible to the hydroxyl group.
- The physical state of lower alcohols are liquids
- Alcohols with more than 12 carbon atoms are solids
- Boiling point of alcohols increases with the increase in molecular mass.
- Lower alcohols are miscible
- Higher alcohols those having six or more than are in soluble in water

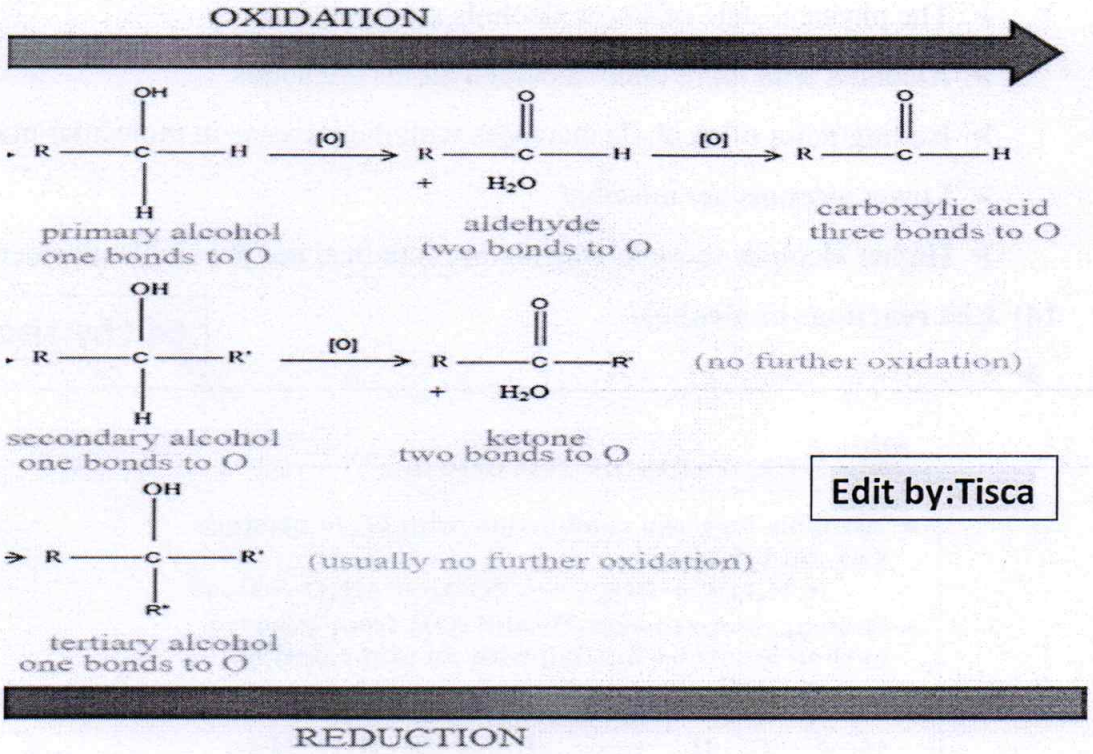
14) List reactions of alcohols

Edit by:Tisca



- Oxidation reaction: primary alcohol are oxidized to **aldehyde (R-CHO)** which are further oxidized to **carboxylic acid (R-COOH)**

- **Secondary alcohols** are oxidized to **ketones**
- **Tertiary alcohols** cannot be oxidized because there is **no hydrogen atom** that bond to the carbon.



Edit by:Tisca

15) Write some common uses of alcohols

- **Methanol:** is used as solvent, antifreezes for car radiators and used as fuel.
- **Ethanol:** used as solvent, manufacture of dyes, drugs, cosmetics, gasoline
- **2-propanol** is used to clean skin before medical procedures& sterilize medical instruments.
- **Glycerol: used as lotions, liquid soap and shaving cream**

16) Define phenols

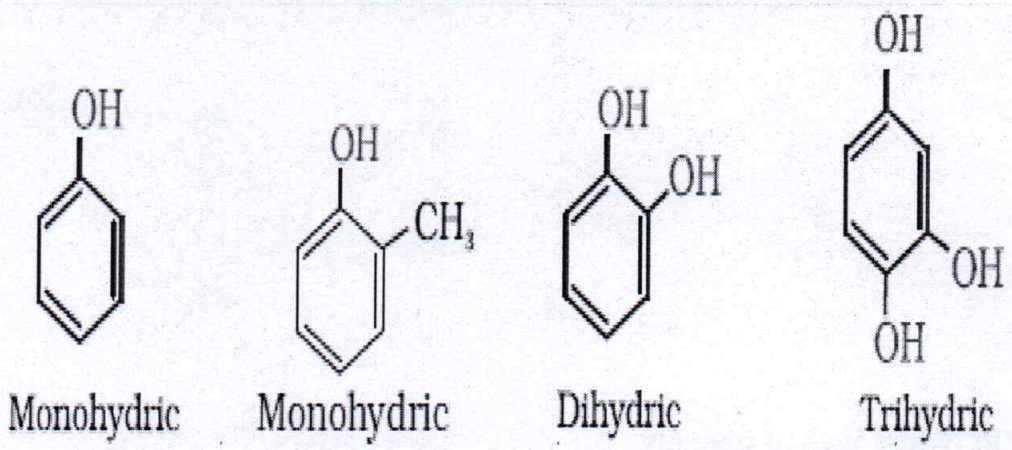
- **Phenols** are compounds that have hydroxyl group bonded directly to the benzene. **Ar-OH** where **Ar** is Aryl or benzene derivatives.

17) According to the hydroxyl group attached. What are phenols be classified?

Edit by:Tisca

Phenols can be classified into three groups

- **Monohydric phenol:** are phenols that contain one hydroxyl group
- **Dihydric phenol:** are phenols that contain two hydroxyl group
- **Tihydric phenol:** are phenols that contain three hydroxyl group

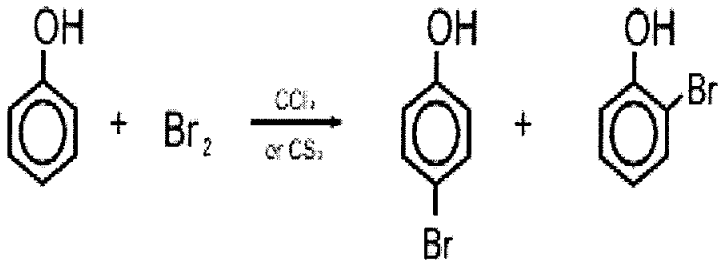


18) Write physical properties of phenols

- Phenols form hydrogen bond with other phenols and water
- Have higher melting and boiling points
- Phenols are more soluble in water

19) List reactions of phenols

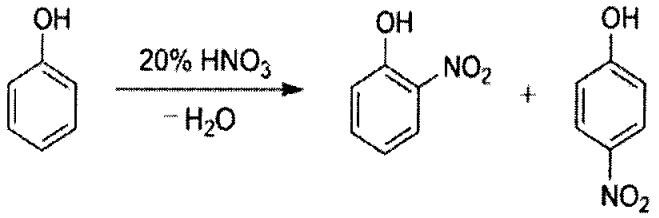
- **Halogenation:** is the addition of halogens



- **Note**: substitution occurs primarily in the position **Para** to the hydroxyl group. When the Para position is blocked **ortho** substitution is observed.

- **Nitration:** is a process of nitrating phenols with dilute nitric acid

Edit by:Tisca



20) Write uses of phenols

- Phenols are used to prepare aspirin , dyes and plastics
- Phenols are used as antiseptic for washing surgical instruments.
- Phenols are also used in perfumery

21) Define Ether

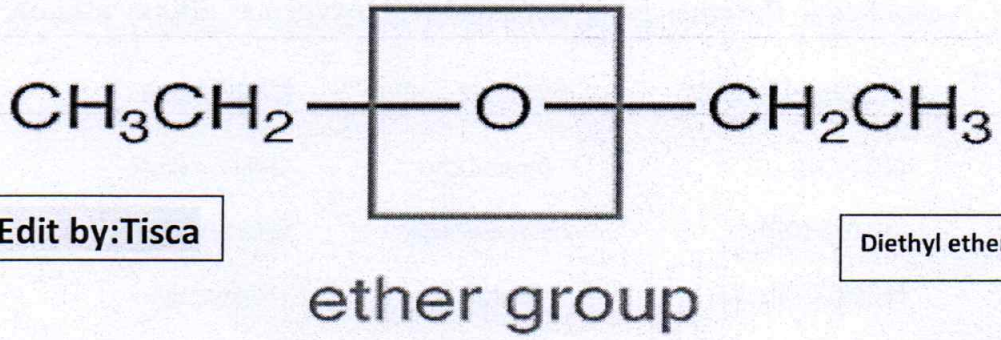
- Ethers are organic compounds which contain **Oxy group (-O-)** as functional group. **R-O-R**

22) On the basis of alkyl or aryl group. What are ethers be classified

- Ethers can be classified **symmetrical** and **unsymmetrical** ethers

23) What is the difference between symmetrical and unsymmetrical ether

- **Symmetrical ether** is simple ether that the alkyl or aryl group attached to the either side of the oxygen atoms is the same.

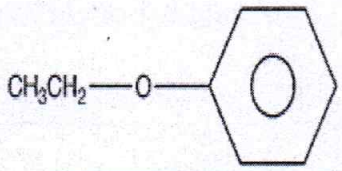


Edit by:Tisca

Diethyl ether

- **Unsymmetrical ether:** is mixed ether in which the alkyl or aryl group attached to the either side of the oxygen atoms are not the same.

Ethyl methyl ether



Ethyl phenyl ether

24) How ethers are named

Ethers can be named by two ways

- **Common system** and **IUPAC system**
  - **In common system** the names of alkyl group are listed in alphabetical order and followed by the word ether and prefix is used if both alkyl groups are same.
  - **In the IUPAC system** ethers are named by choosing larger alkyl group attached to the ether group as parent while the smaller alkyl group attached to the ether group is named with oxygen as **alkoxy alkane. E.g.**

Structural formula	Common name	IUPAC name
CH3-O-CH3	Dimethyl ether	Methoxymethane
CH3-O-CH2CH3	Ethyl mehtyl ether	Methoxyethane
CH3CH2-O-CH2CH3	Diethyl ether	Ethoxyethane

Edit by:Tisca

25) What is the isomerism in ethers?

- Ethers exhibit **functional isomerism** with alcohols having same number of carbon atoms but differ in functional group.

CH_3-CH_2-OH — Ethanol

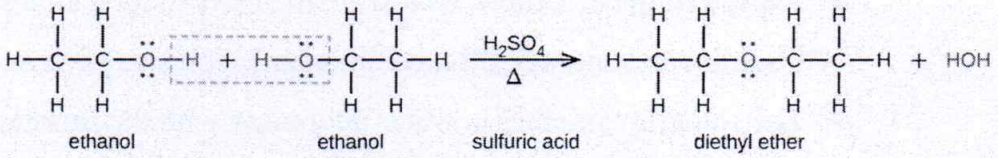
Both have C_2H_6O

H_3C-O-CH_3 — Dimethyl Ether

26) How Ethers are prepared?

Ethers are prepared by

- **Dehydration of alcohol** in this process two molecules of primary alcohol react by dehydration to produce symmetrical ether.



- **Dehalogenation of haloalkanes** in this process ethers are prepared from alkyl halides and sodium alkoxide via substitution reaction known as **William synthesis.**

Edit by:Tisca

$R\text{-}O\text{-}Na^+ + X\text{–}R$ ----------$R\text{-}O\text{-}R + NaX$

Sodium alkoxide + alkyl halide ---------- Ether + NaX

E.g: Sodium ethoxide + bromo methane ---------- ethyl mehyl ether + sodium bromide

$CH_3CH_2ONa + Br\text{-}CH_3$---------- $CH_3CH_2\text{-}O\text{-}CH_3 + NaBr$

27) **List physical properties of ethers**

- **Dimethyl ether** and **ethyl methyl ether** are gases while other ethers are liquids.
- **The boiling point** of ethers is higher than hydrocarbons and lower than alcohols.
- **Alkyl groups of Ethers** with more than five carbon atoms are insoluble in water but dissolve in organic solvents such as benzene.
- The solubility of ether in water **decreases** with the **increase** in the molecular mass.

Edit by:Tisca

28) **List reaction of ethers**

Ethers are not very reactive however under suitable condition ethers undergo:

- **Combustion reaction:** in this process ethers burn in air to produce carbon dioxide and water.

$$C_2H_5\text{-}O\text{-}C_2H_5 + 6O_2 \longrightarrow 4CO_2 + 5H2O$$

- **Dehydration:** In this process ether vapors are passed over Alumina (Al_2O_3) to produce **alkene.**

$$C_2H_5\text{-}O\text{-}C_2H_5 \longrightarrow 2CH_2{=}CH_2 + H_2O$$

29) **List uses of ethers**

- **Ethers** are widely used as solvents for oils , gums and resins
- **Diethyl ether** is used as general anesthetic in surgery but it is usage for anesthetic is diminishing as better anesthetic are now available.

CHEMISTRY FORM FOUR

QUESTIONS AND ANSWERS

CHAPTER Three

ALDEHYDES, KETONES AND CARBOXYLIC ACIDS

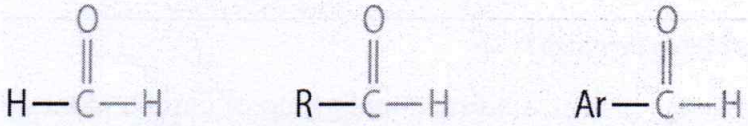
1. Define aldehyde

✓ **Aldehyde** are organic compounds which contain aldehydic carbonyl (-CHO) as functional group.

2. What is the general formula of aldehyde?

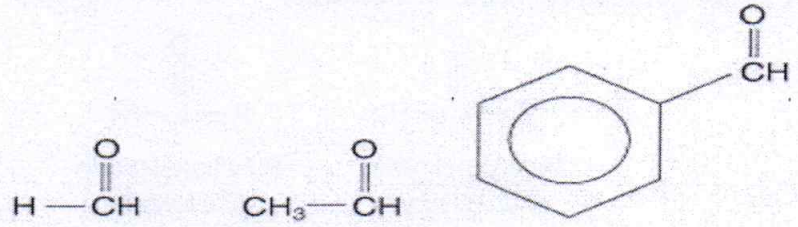
✓ Aldehyde has the general formula of

By: Tisca



R=alkyl group **Ar**-Aryl group or hydrogen

✓ **Aryl group** derived from an aromatic hydrocarbon by the removal of a hydrogen atom, as phenyl-C_6H_5



methanal (formaldehyde) ethanal (acetaldehyde) benzaldehyde

3. **How Aldehydes are named and give examples ?**

   ✓ Aldehyde are named by using **Common** and **IUPAC system**

I.U.P.A.C. Name	Common Name	Formula
Methanal	Formaldehyde	$H-\overset{\overset{O}{\|\|}}{C}-H$
Ethanal	Acetaldehyde	$CH_3\overset{\overset{O}{\|\|}}{C}-H$
Propanal	Propionaldehyde	$CH_3CH_2\overset{\overset{O}{\|\|}}{C}-H$
Butanal	Butyraldehyde	$CH_3CH_2CH_2\overset{\overset{O}{\|\|}}{C}-H$
Pentanal	Valeraldehyde	$CH_3CH_2CH_2CH_2\overset{\overset{O}{\|\|}}{C}-H$

4. **What is a carbonyl group?**

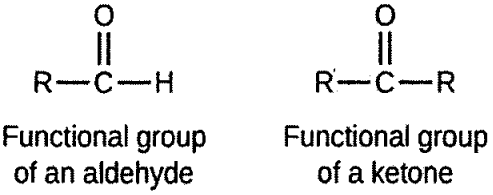
   ✓ **Carbonyl group** is a functional group of carbon atom double bonded to an oxygen atom (C=O).

By: Tisca

5. **What are ketones?**

   ✓ **Ketones** are organic compounds which contain carbonyl group that are boned to two alkyl groups.

6. **What is the general formula of ketones?**



Functional group of an aldehyde

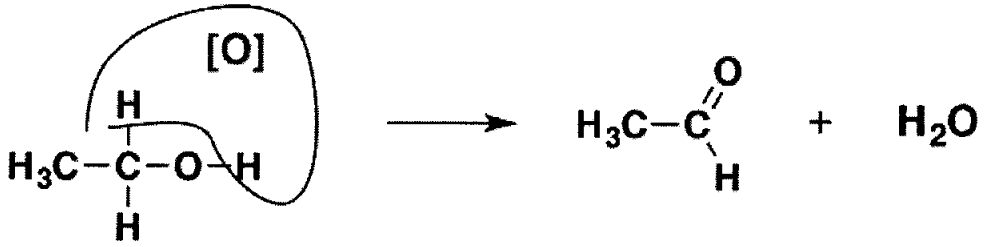
Functional group of a ketone

7. **What are the characteristics of ketones and Aldehyde?**
   - ✓ **Aldehyde** and **ketones** have a polar carbonyl group; therefore they are polar molecules with strong intermolecular force than the hydrocarbons.
   - ✓ **Aldehyde** and **ketones** have no oxygen hydrogen bond **(O-H),** therefore two molecule of aldehyde or ketones are incapable of intermolecular hydrogen bonding.
   - ✓ **Aldehyde** and **ketones** have higher boiling points than hydrocarbons of similar mass.
   - ✓ **Aldehyde** and **ketones** have lower boiling points than alcohols of similar mass because of not having hydrogen bond between them.
8. **What is the difference b/w aldehyde and ketones?**
   - ✓ **Aldehyde** has at last one hydrogen atom bonded to the carbonyl or if the carbonyl group is primary carbon the molecule is an aldehyde.

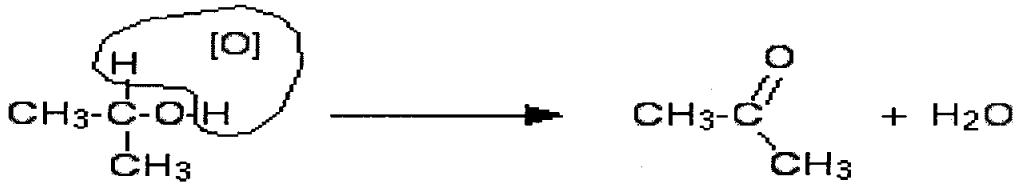
     By: Tisca
   - ✓ **Ketones** have two alkyl groups boned to the carbonyl group or if the carbonyl group is secondary the molecule is ketones.
9. **Why ketones and aldehyde are called carbonyl compounds?**
   - ✓ Because they posses **carbonyl group (C=O)** that is why they are called carbonyl compounds.

10.How Aldehydes and ketones are produced?

✓ **Primary alcohols** are oxidized to produce aldehyde.



✓ **Secondary alcohols** are oxidized to produce ketones.

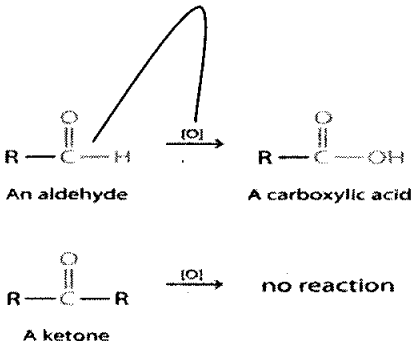


propan-2-ol propanone

By: Tisca

11.What is the product by the oxidation of aldehyde?

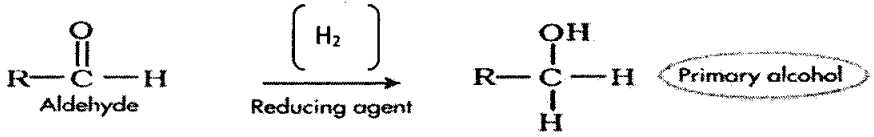
Aldehydes can be oxidized to **carboxylic acid** while **ketones** cannot be oxidized.



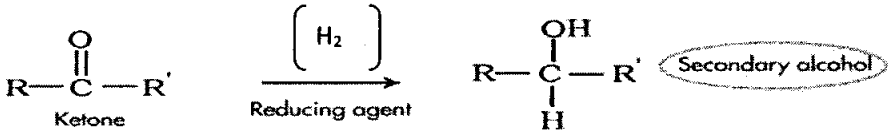
12. What is the production formed by the reduction of aldehyde and ketone?

- ✓ **Aldehydes are** reduced to primary alcohol (1^0) by the process of catalytic hydrogenation.
- ✓ **Ketones** are reduced to secondary alcohol (2^0) by using the process of catalytic hydrogenation.

Reduction of aldehydes and ketones



By: Tisca



13. List uses of aldehyde and ketones

- ✓ **In aldehyde: Formaldehyde: $CH_2=O$** is the simplest aldehyde which is used for the synthesize of many resins like urea formaldehyde and plastics such as Bakelite.
- ✓ **Formalin**: is an aqueous solution of formaldehyde that is used as a disinfectant and preservative for biological specimen.
- ✓ **Acetaldehyde** is used the production of acetic acid
- ✓ **Vanillin** is used as a fragrance and in food flavoring.
  - **In ketones: Acetone:** RCOR is the simplest ketone that is used industrial solvent and the formation of organic polymers.

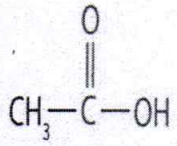
14. Define carboxylic acid

- ✓ **Carboxylic acids** are organic compounds that contain carboxyl (-COOH) as a functional group.

15. According to the number of carboxyl group. What are classified carboxylic acids?

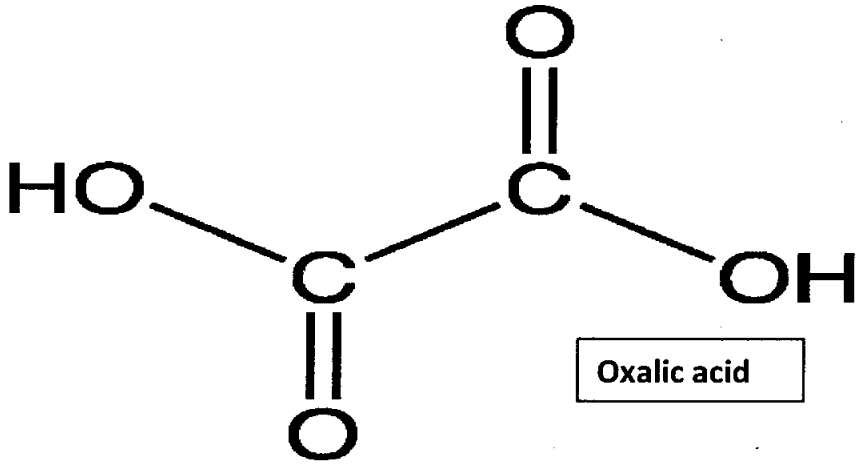
By: Tisca

- ✓ **Mono-carboxylic acid:** are carboxylic acids that contain one carboxyl group in the molecule.



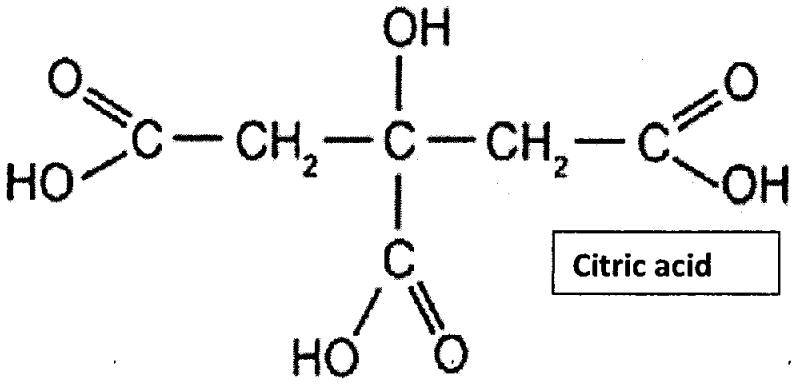
Acetic acid

- ✓ **Di-carboxylic acid**: are carboxylic acids that contain two carboxyl groups in their molecules.



Oxalic acid

- ✓ **Tri-carboxylic acid:** are carboxylic acids that contain three carboxyl groups in their molecules.



Citric acid

By: Tisca

16. How carboxylic acids are named and give examples?

✓ **Carboxylic acids** are named by both common system and IUPAC system

Length of Carbon Chain	Structural Formula	Common Name[a]	IUPAC Name
C_1 monoacid	H—COOH	formic acid	methanoic acid
C_2 monoacid	CH_3—COOH	acetic acid	ethanoic acid
C_3 monoacid	CH_3—CH_2—COOH	propionic acid	propanoic acid
C_4 monoacid	CH_3—$(CH_2)_2$—COOH	butyric acid	butanoic acid
C_5 monoacid	CH_3—$(CH_2)_3$—COOH	valeric acid	pentanoic acid
C_6 monoacid	CH_3—$(CH_2)_4$—COOH	caproic acid	hexanoic acid

[a] The mnemonic "*F*rogs *a*re *p*olite, *b*eing *v*ery *c*ourteous" is helpful in remembering, in order, the first letters of the common names of these six simple saturated monocarboxylic acids.

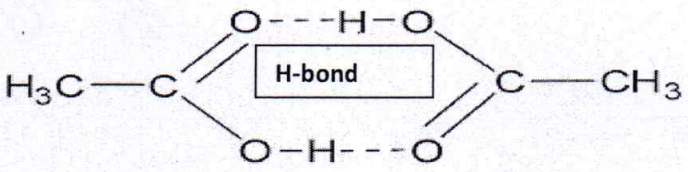
17. List physical properties of carboxylic acids

✓ **Carboxylic acids** possess a hydrogen atom bonded to an electronegative oxygen atom.

By: Tisca

✓ **Carboxylic acids** are often existing as dimmers, i.e.: structure containing two identical or similar units held together by two intermolecular hydrogen bonds.

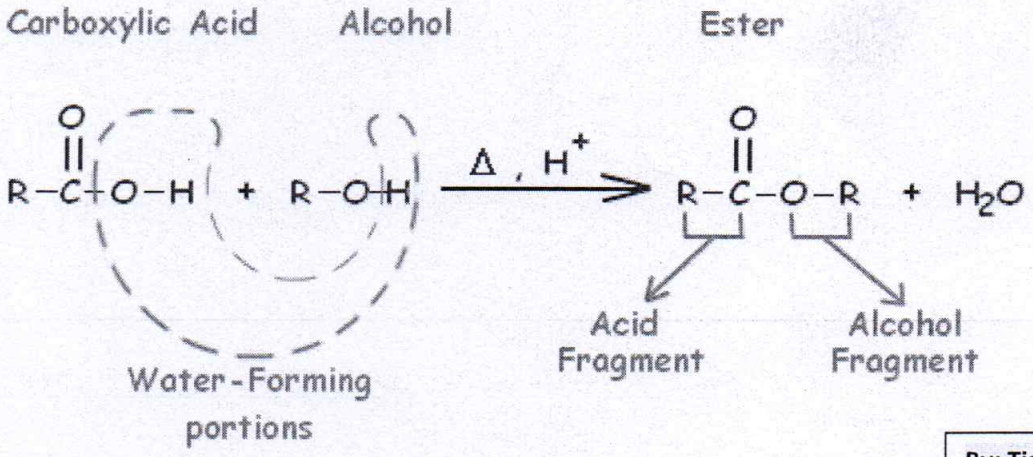
✓ **Carboxylic acids** have higher boiling points and melting points than alcohols of similar mass.



Dimerization of acetic acid

18. What is esterification?

- ✓ **The reaction** of **alcohol** and **carboxylic acid** to form **ester** and **water** is known as **esterification.**



By: Tisca

19. List uses of carboxylic acids

- ✓ **Acetic acid** is the sour tasting component of vinegar and also used industrially as reagent for the synthesis of esters and manufacture of plastics.
- ✓ **Aspirin (acetylsalicylic acid**) is a pain reliever and anti-inflammatory agent.
- ✓ **Adipic acid:** is used to make nylon 6.6
- ✓ **Fatty acids**: are main components of fats and oils.

By: Tisca

CHEMISTRYFORM FOUR

QUESTION AND ANSWERS

CHAPTER FOUR BIO-CHEMISTRY

Part one: Answers

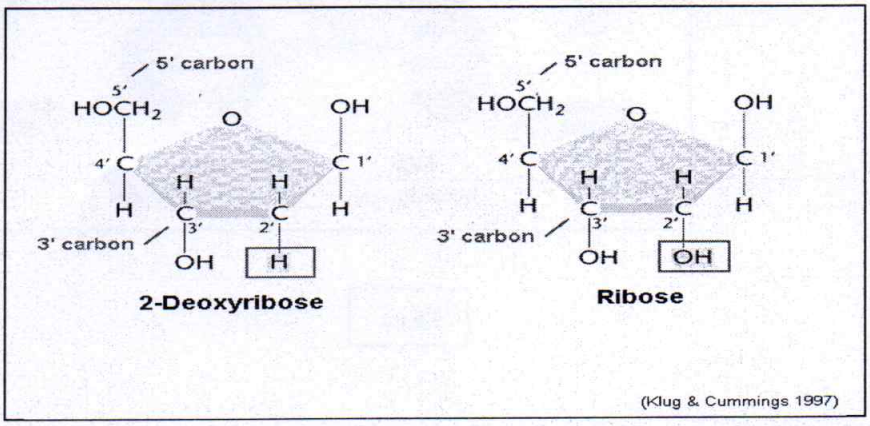
1. **Define a) Carbohydrate b) nucleic acid c) proteins d) lipids**

- ✓ **carbohydrates** are poly-hydroxyl aldehydes or ketones
- ✓ **nucleic acids** are poly nucleotides that occur in living cells.
- ✓ **proteins** are polymers of amino acids connected by peptide bond.
- ✓ **lipids** are organic chemical compounds that are insoluble in water but soluble in non-polar solvents.

Tisca

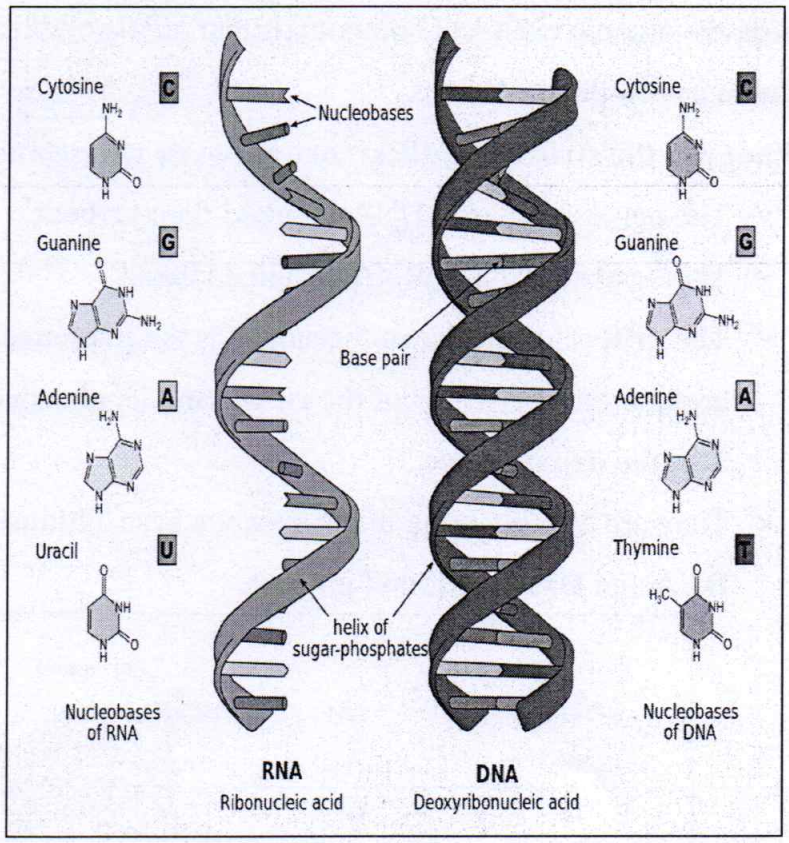
2. **What are the structural difference between ribose and deoxyribose?**

- ✓ The pentose sugar in DNA is called **deoxyribose**.
- ✓ The pentose sugar in RNA is called **ribose**.
- ✓ The **difference between** the sugars is the **presence** of the **hydroxyl** group on the carbon 2' of the **ribose** and its **absence** on the Carbon 2' of the **deoxyribose.**
- ✓ They are also differing in nitrogenous base **Thiamine** is found in **DNA** and **Uracil** is found in **RNA**



3. **What are the structural difference between DNA and RNA?**

- ✓ The sugar present is **RNA** is **ribose.**
- ✓ The sugar in **DNA** is **deoxyribonucleic acid.**
- ✓ The nitrogenous base in **RNA** is **adenine, guanine, cytosine and uracil**
- ✓ The nitrogenous base in **DNA** is **adenine, thymine, guanine, cytosine**
- ✓ RNA has **single helix** while DNA has **double helix**



Tisca

4. **Why some amino acids are essential?**

✓ **Essential amino acids** cannot be made by the body.

They're **needed** for **vital** processes like the building of proteins and synthesis of hormones and neurotransmitters

5. **State two biological importance of proteins**

Proteins have **structura**l and **functiona**l importance including:

✓ **Keratin:** gives strength and support to tissues. Hairs, horns, hooves and fingernails are all made up of keratin.

✓ **Insulin:** is a hormone that regulates blood glucose level

✓ **Hemoglobin:** transport oxygen and carbon dioxide from lungs to tissues and from tissues to the lungs **Tisca**

6. **What is a protein denaturation? Name denaturating agents?**

✓ **Denaturation** is a process that disrupts secondary, tertiary and quaternary structure. Agents of denaturation include

✓ Heat, mechanical agitation, organic compounds, Changes in PH and heavy metals

7. **Identify and name the following tripeptide**

✓ **Glycyl-glycyl-glycine** is a tripeptide in which three glycine units are linked by peptide bonds in a linear sequence.

8. **Give an example of each type of lipid**

✓ **Monounsaturated fatty acid:** include linolenic acid, oleic acid, elaidic acid and vacentic acids

✓ **Saturated triacylglycerol:** butyric acid, palmitic acid, and stearic acid.

9. What is the function of nucleic acids?

Nucleic acids are essential for life and play a variety of roles such as:

- ✓ They control protein synthesize
- ✓ Store and transmit genetic information from one generation to the next.

10. What molecules do both DNA and RNA contain?

- ✓ Both **DNA** and **RNA** contain **nucleotides** which are **pentose sugar**, **Nitrogenous base** and **phosphate group**.

11. List Characteristics and importance of lipids

- ✓ **Lipids** are organic chemical compounds that are insoluble in water but soluble in non-polar solvents. **Tisca**
- ✓ **Lipids** act as a source of energy and are the natural solvent for fat soluble vitamins.
- ✓ They have a role in **protection** and **fixation of internal organs.**
- ✓ Have myline sheath and serve as **electrical insulator.**
- ✓ Lipids under the skin serve as **thermal insulator**.

12. What are waxes?

- ✓ **Waxes** are soft solids with low melting points which are insoluble in water.

13. List types of phospholipids

On the basis of the type of alcohol phospholipid are two types

- ✓ Glycerophospholipid:
- ✓ Sphingolipids

14. **What is the difference between saturated and monounsaturated fatty acids?**

✓ Fatty acids that have no double bon is a **saturated fatty acid** while fatty acids that have one or more double bond is referred to as **unsaturated.**

15. **Define amino acids**

- ✓ **Amino acids** are monomers of proteins.
- ✓ **amino acids** are the building blocks of proteins

16. **Define primary structure of proteins**

- ✓ **Primary structure of proteins** is the sequence of amino acids that is joined together by peptide bond.

Tisca

17. **According to the molecular shape. What are proteins be classified?**

- ✓ **Globular proteins:** are compact, spherical structures that are soluble in aqueous environment.
- ✓ **Fibrous proteins:** are long thread like structures that have high helical content.

18. **State two most common types of secondary structure of proteins**

- ✓ Alpha helix:
- ✓ Beta-pleated sheet

19. **Classify amino acids based on the relative number of amino and carbonyl groups in their molecule?**

- ✓ Equal number of amino and carbonyl group makes **neutral**
- ✓ More number of amino than carbonyl groups makes **basic**
- ✓ More carbonyl groups than amino groups make **acidic**

20. List types of amino acids

✓ **Essential amino acids:** amino acids that are synthesized in the body is known as non-essential amino acids. E.g.

Essential amino acids

✓ Arginine	Lysine
✓ Histidine	Methionine
✓ Isoleucine	Threonine
✓ Leucine	Phenylalanine
✓ Valine	Tryptophan

✓ **Nonessential amino acids:** amino acids that are not synthesized in the body but obtained from the diet is known as essential amino acids.

Non-Essential amino acids

✓ Alanine	Glycine
✓ Asparagine	Proline
✓ Aspartate	Serine
✓ Glutamate	Cysteine
✓ Glutamine	Tyrosine

Tisca

21. What are functions of carbohydrates?

✓ **Carbohydrates** act as a source of energy. They are regarded as body fuel.

✓ **Carbohydrates** act as food reserve in plants as **starch** and in animals as **glycogen**.

✓ **Carbohydrates** are the structure material for the cell wall such as **cellulose**

22. **What is the difference between?**

a) **Sugars** and **non-sugars**

b) **Reducing sugar** and **non- reducing sugars.**

c) Monosaccharide, disaccharide and polysaccharide

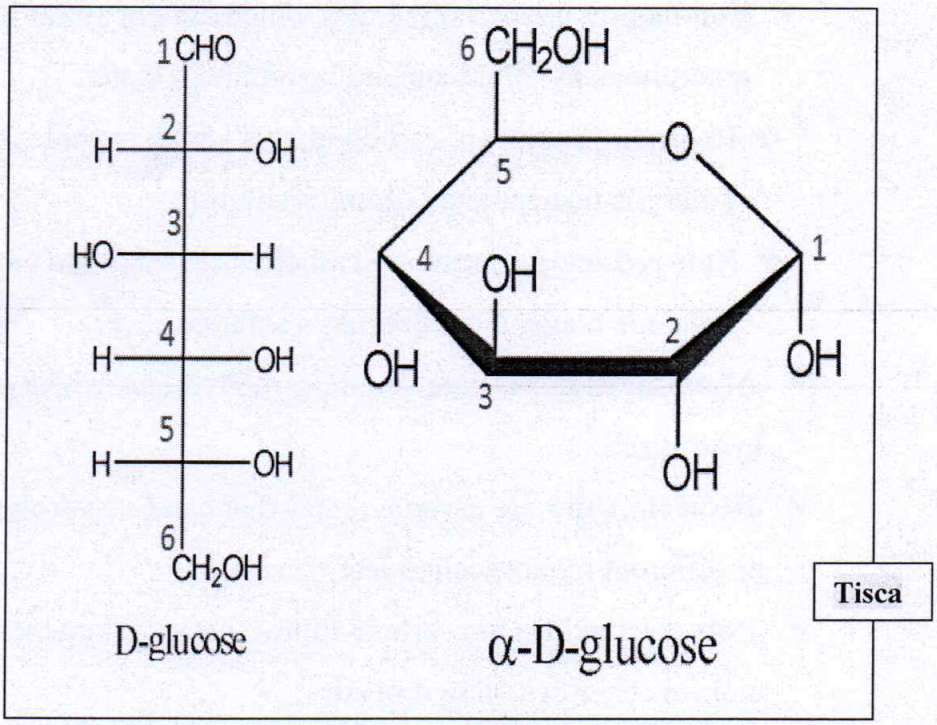
- ✓ **Sugars** are carbohydrates are sweet in taste, have crystalline in nature and soluble in water.
- ✓ **Non-sugars** are carbohydrates which are not sweat in taste, have amorphous in nature and are insoluble in water.
- ❖ **Reducing sugar** are carbohydrates which capable of reducing Tolle n's reagent and Fehling's solution.
- ❖ **Non-reducing sugars** are carbohydrates that are unable to reduce Tolle n's reagent and Fehling's solution.

Tisca

- ✓ **Monosaccharides:** are simple carbohydrates which cannot be hydrolyzed.
- ✓ **Disaccharides** are carbohydrates that can be hydrolyzed in to same or different monosaccharides.
- ✓ **Polysaccharides** are carbohydrates that give large number of monosaccharides on hydrolysis

23. Differentiate Fisher projection and Haworth projection

- ✓ **Fisher projection** is a formula that represents the straight chain of monosaccharides. The chiral carbon atoms are intersection of horizontal and vertical lines.
- ✓ **Haworth projection** is the formula that represents the cyclic structure of monosaccharides.



24. According to the functional group. What are monosaccharides be classified?

- **Aldose sugar** which contain aldehydes carbonyl group.
- **Ketose sugar** which contain ketone functional group.

25. What is biochemistry?

- ✓ Biochemistry is the study of biomolecules.

26…Differentiate metabolism and biomolecules?

- ✓ Any chemical process that occur in living system is known as **metabolism**
- ✓ The chemical constituents of the living system are known as **biomolecules**

27… list major types of biomolecules

Biomolecules are mainly classified as

- ❖ Carbohydrates
- ❖ Proteins
- ❖ Lipids
- ❖ Nucleic acids

28. what are the fields of biochemistry?

Biochemistry is divided into three fields

- ❖ **Structural biochemistry:** is the study of structure of biomolecules
- ❖ **Enzymology:** is a branch of biochemistry concerned with enzymes.
- ❖ **Metabolism:** is the sum of all chemical reactions in the organism.

Tisca

29…On the basis of number of carbon.

What are monosaccharides be classified?

- ✓ **Triose:** Having three carbon atoms = $C_3H_6O_3$
- ✓ **Tetroses:** having four carbon atoms = $C_4H_8O_4$
- ✓ **Pentose:** Having five carbon atoms = $C_5H_{10}O_5$
- ✓ **Hexose:** Having six carbon atoms = $C_6H_{12}O_6$

30.. Differentiate pyranose and Furanose structure

- ✓ **Pyranose structure** is a six membered rings containing an oxygen atom.
- ✓ **Furanose structure** is five membered rings containing an oxygen atom

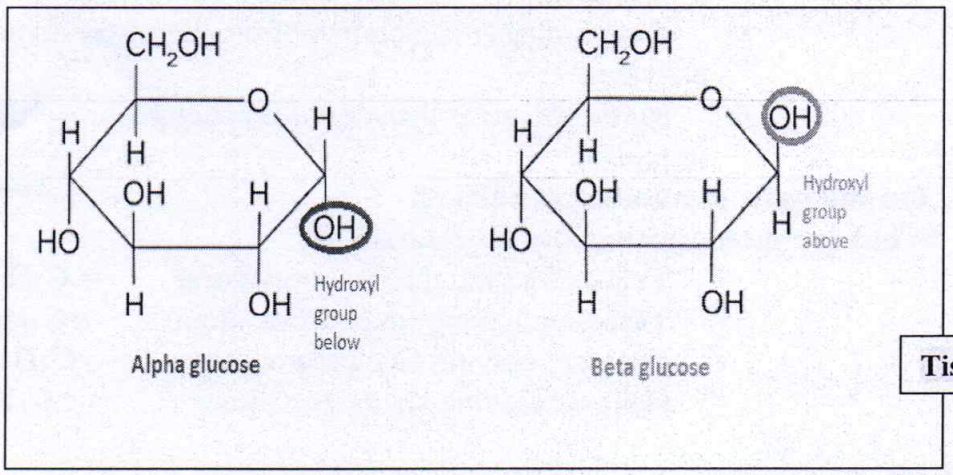
31. What are lipids be classified?

- ❖ **Simple lipids** are esters of fatty acids and alcohol. They are dived as **Triglycerides** and **waxes.**

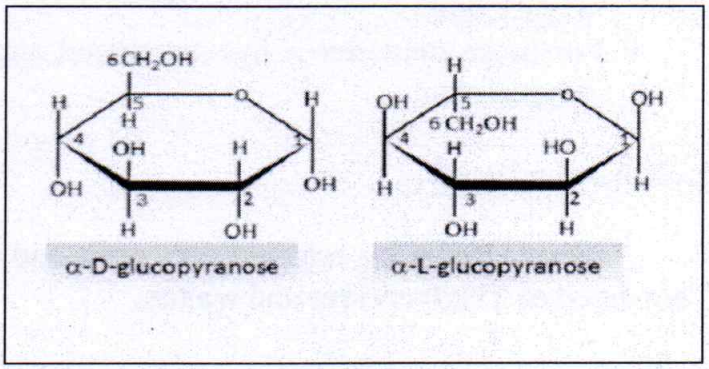
- **Compound lipids** are esters of fatty acids with glycerol and additional group. They are classified as
  - ✓ **Glycolipid**: is a combination of lipid and carbohydrates
  - ✓ **Lipoproteins**: is a combination of lipid and proteins
  - ✓ **Phospholipids** is a combination of a phosphate group and amino alcohol.
- **Derived lipids** are substance derived by the hydrolysis of compound and simple lipids.

32. Differentiate Alpha glucose and beta glucose

- ✓ If the –**OH** attached to the **anomeric carbon** is below the ring the sugar is called **alpha sugar.**
- ✓ If the **hydroxyl group** on the **anomeric carbon** is above the sugar is beta sugar.



Note: The Sugar is D Sugar if the CH_2-OH is above the ring and L sugar if the CH_2-OH is in or below the ring.

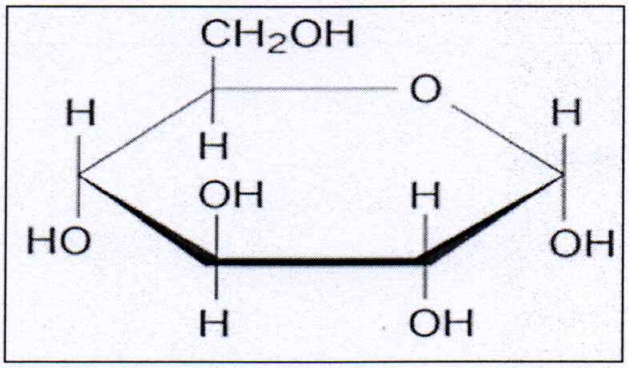


Part two: Multiple choice

1. **Which of the following is a polysaccharide?**
   - a) **Cellulose**
   - b) Fructose
   - c) Glucose
   - d) Sucrose
2. **What elements make up carbohydrates**
   - a) Carbon, hydrogen, nitrogen, oxygen
   - b) Sulfur, carbon, hydrogen
   - c) Glucose and fructose
   - d) **Carbon, hydrogen, oxygen**
3. **$C_6H_{12}O_6$ is the molecular formula for what carbohydrate**
   - a) Sucrose
   - b) glucose
   - c) Fructose
   - d) **Both glucose and fructose**

Tisca

4. **Which biomolecule is this picture**



   - a) Protein
   - b) Nucleic acid
   - c) Lipid
   - d) **Carbohydrate**

5. **The information of protein synthesis is stored in**
   a) Amino acids
   b) Fats
   c) Glucose
   d) **Nucleic acids**
6. **The repeating units in both DNA and RNA are called**
   a) Amino acids
   b) Hydrogen bond
   c) Nucleic acids
   d) **Nucleotides**
7. **The sugar that is in the backbone of DNA is called**
   a) Delta glucose
   b) **Deoxyribose**
   c) Fructose
   d) Ribose
8. **The sugar in the backbone of RNA is called**
   a) Deoxyribose
   b) Fructose
   c) **Ribose**
   d) Sucrose

Tisca

9. **The RNA molecule is a**
   a) Double helix
   b) Double strand molecule
   c) None of the choices
   d) **Single stranded molecule**
10. **Adenine bonds with**
   a) Guanine
   b) **Thymine**
   c) Cytosine e
   d) Adenine
11. **Guanine bonds with**
   a) Adenine
   b) Guanine
   c) **Cytosine**
   d) Thymine

12. **Nucleotides are made up of a sugar, phosphate and**
    - a) **Nitrogenous base**
    - b) Nucleic acid
    - c) Cholesterol
    - d) Fatty acid
13. **Which of the following is monosaccharide?**
    - a) **Fructose**
    - b) Glycogen
    - c) Lactose
    - d) Sucrose
14. **Which component is found in sphingolipids?**
    - a) A carbohydrate
    - b) A negative charge
    - c) A phosphate group
    - d) **Amino alcohol**
15. **Which of these Is not a lipid**
    - a) Fats
    - b) Oils
    - c) **Proteins**
    - d) Waxes

Tisca

16. **What are functions of lipids**
    - a) Provide energy to the body
    - b) Form part of cellular membrane
    - c) Control metabolism
    - d) **All options given are correct**

CHEMISTRY FORM FOUR

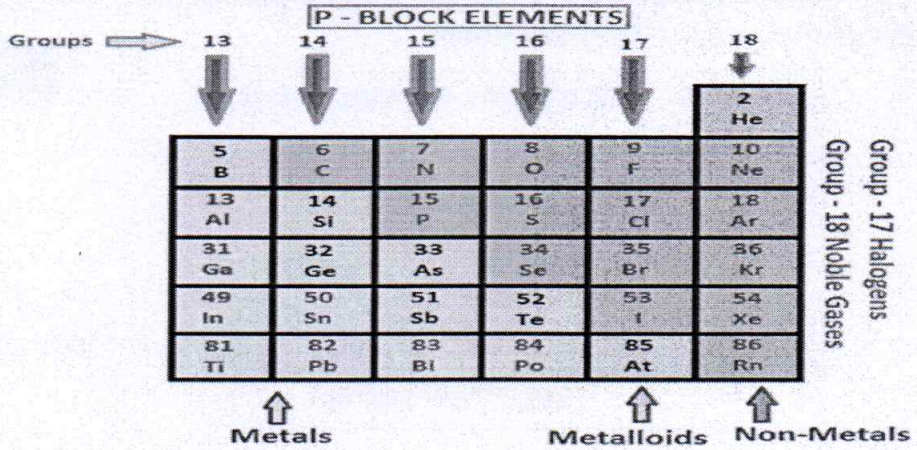
QUESTION AND ANSWERS

CHAPTER FIVE P-BLOCK ELEMNTS

1) **Define p-block elements**
   - **P-block elements** are those in which the last electron enters the outermost p-orbital
2) **What is the total number of p-block elements?**
   - There are **six groups of p-block elements** in the periodic table from group 13 up group 18.
3) **List groups of p-block elements**
   - **Group 13 or group IIIA** is called Boron family
   - **Group 14 or group IVA** is called Carbon family
   - **Group 15 or group VA** is called Nitrogen family
   - **Group 16 or group VIA** is called Oxygen family
   - **Group 17 or group VIIA** is called Halogens
   - **Group 18 or group VIIIA** is called Noble gases

Tisca

- **Why they are called p-block elements?**
  - because their valence electrons are entering **p-orbital**.
- **Classify p-block elements into metals, non-metals and metalloids**



4) **Define Boron family**
   - **Group 13 elements** of the periodic table are called **boron family.**

5) **List Boron family** **symbols**
   - Boron= B
   - Aluminum= Al
   - Gallium= Ga
   - Indium= In
   - Thallium= Ti

6) **What is Boron?**
   - **Boron** is the lightest of the elements in this group. **It is a metalloid**. The others in this group **are metals**.

7) **Identify the position of boron family in the periodic table**
   - **Boron family** is located the **group 13** in the periodic table.

8) **Write at least three physical properties of boron family**
   - In nature boron is a **metalloid** while the rest in the groups **are metals**.
   - At room temperature the **physical state** of boron family is **solid**
   - Aluminum is **amphoteric,** therefore it can react with both **acids** and **bases**
   - Boron has a **highest melting point** while gallium has the **lowest melting points**.

9) **State at least three common uses of boron family**
   - **Boron** is used in manufacturing of Pyrex and making bullet proof vest
   - **Aluminum** is used for making cooking utensils and window frames.
   - **Indium** is used in making semi-conductor devices such as transistors.
   - **Thallium** is extremely toxic and has no commercial use

10) **Write electron configuration of Boron family**

Tisca

Element	Electronic configuration (ns^2np^1)
$_5B$	$[He]2s^22p^1$
$_{13}Al$	$[Ne]3s^23p^1$
$_{31}Ga$	$[Ar]3d^{10}4s^24p^1$
$_{49}In$	$[Kr]4d^{10}5s^25p^1$
$_{81}Tl$	$[Xe]4f^{14}5d^{10}6s^26p^1$

11) What is noble gas core notation?

- **Noble gas core notation** is an abbreviation in an atom's electron configuration.

12) Define carbon family

- **Group 14 elements** of the periodic table are called carbon family.

13) List members of carbon family **symbol**

- Carbon= C
- Silicon= Si
- Germanium = Ge
- Tin (Stanum) = Sn
- Lead (Plumbum)= Pb

Tisca

14) Where is the position of carbon family in the periodic table?

- **Carbon family** is found **group 14** in the periodic table

15) Write at least three physical properties of carbon family

- In nature carbon is **non-metal**, silicon and germanium are **metalloids** Tin and Lead are **metals.**
- At room temperature the physical state of all members of carbon **family are solids**.
- The melting and boiling points of **group 14 elements** are much higher than those of corresponding elements **of group 13.**

16) State at least three common uses of carbon family

- **Carbon** is used in pencils and electrochemical cells in the form of graphite.
- Since **silicon and germanium** are metalloids, they are used in electronics and semi-conductors.
- **Lead** is used to protect your torso during dental X-ray

17) Write the electron configuration of carbon family

Element	Atomic Number	Electronic Configuration
Carbon	6	$[He]\ 2s^2\ 2p^2$
Silicon	14	$[Ne]\ 3s^2\ 3p^2$
Germanium	32	$[Ar]\ 3d^{10}\ 4s^2\ 4p^2$
Tin	50	$[Kr]\ 4d^{10}\ 5s^2\ 5p^2$
Lead	82	$[Xe]\ 4f^{14}\ 5d^{10}\ 6s^2\ 6p^2$

The general valence shell of group 14 **is ns^2np^2**

18) Define nitrogen family

- **Group 15** or VA elements of the periodic table are called **nitrogen family**.

19) Where is the position of nitrogen family in the periodic table?

- The nitrogen family is located in the **p-block in Group 15**

20) Write three physical properties of nitrogen family

- **In nature nitrogen** and phosphorous are **non-metals** while arsenic, bismuth and antimony are **metals**.
- **All nitrogen family** show **allotropes** except nitrogen and bismuth
- **Nitrogen exist as diatomic** gas while others exist as tetra-atomic solids

21) State common uses of nitrogen family

- **Nitrogen** and **phosphorous** are used for fertilizers and essential for life.
- **Arsenic** which is notorious for it is toxicity used in agriculture to control insects and pests
- **Antimony** and **bismuth** are used chiefly in metal alloys.

22) Why nitrogen family are called pnictogens?

- Because all elements in this group are suffocating in nature

23) List members of nitrogen family **Symbol**

- Nitrogen N
- Phosphorous P
- Arsenic As
- Antimony (Stibium) Sb
- Bismuth Bi

Tisca

24) Write electron configuration of nitrogen family

Elements	Electronic configuration ($ns^2\ np^3$)
$_{7}N$	$[He]2s^2\ 2p^3$
$_{15}P$	$[Ne]3s^2\ 3p^3$
$_{33}As$	$[Ar]3d^{10}4s^2 4p^3$
$_{51}Sb$	$[Kr]4d^{10}5s^2 5p^3$
$_{83}Bi$	$[Xe]4f^{14}5d^{10}6s^2 6p^3$

25) Define oxygen family

- **Group 16 or VIA** elements of the periodic table are called oxygen family

26) Where is the position of oxygen family in the periodic table?

- **Oxygen family** are located group 16 or VIA on the periodic table

27) Write at least three physical properties of oxygen family

- In nature oxygen, sulfur and selenium are **non-metals**, tellurium and polonium are **metalloids**
- At room temperature **oxygen is a gas** while other members of the group are **solids**
- **Group 16 elements** show **negative oxidation state of -2** and **positive oxidation** state of **+2, +3, +4 and +6.**

Note: All oxygen family have **positive oxidation state** except **Oxygen** which exhibit +2 oxidation state only in OF_2

28) State at least three uses of oxygen family

- **Oxygen** is essential for respiration in animals and humans.
- **Ozone (O_3)** protects the atmosphere from ultraviolet light
- **Selenium** is widely used in electronic equipment, in pigments and production of glass.

29) Why oxygen family are called chalcogen?

- Because most of the ores in the earth's crust are either **oxides or sulfides**

30) Define ores

- **Ores** are minerals from which metals are extracted.

31) List members of oxygen family	**Symbol**
Oxygen	O
Sulphur	S
Selenium	Se
Tellurium	Te
Polonium	Po

Tisca

32) Write electronic configuration of oxygen family

Gr.16: Oxygen Family/ Chalcogen Family			*(ns^2 np^4) Valence Shell ē: 6*
Elements	**Symbol**	**Atomic No.**	**Electronic Configuration**
Oxygen	O	8	$_2[He]2s^2\ 2p^4$
Sulphur	S	16	$_{10}[Ne]\ 3s^2\ 3p^4$
Selenium	Se	34	$_{18}[Ar]\ 3d^{10}\ 4s^2\ 4p^4$
Tellurium	Te	52	$_{36}[Kr]\ 4d^{10}\ 5s^2\ 5p^4$
Polonium	Po	84	$_{54}[Xe]\ 4f^{14}\ 5d^{10}\ 6s^2\ 6p^4$

33) Define halogens

- **Group 7 or VIIA** elements in the periodic table are called halogens

34) What is the position of halogens in the periodic table?

- The **halogen** elements are found in Group 17 of the **periodic table**

35) Write at least three physical properties of halogens

Tisca

- **Group 17 elements** are poor conductors of electricity
- All group 17 elements are **non-metals** therefore they are insulators of heat and electricity.
- They are poor conductor of heat because they are made up of **covalent bond.**

36) State three common uses of halogens

- **Chlorine** is used as bleaching agent in paper and industry.
- **Halogens like chlorine** are used in purification of water, sanitation, pharmaceutical, bleaching, pesticides and antiseptic.
- **Sodium chloride** and **sodium iodide** are essential dietary minerals for human beings.

37) Why group 17 are called halogens?

- The word **halogens** are meant by **salt makers**, they make salt with metals.

38) List members of halogens **Symbols**

- Fluorine F
- Chlorine Cl
- Bromine Br
- Iodine I
- Astatine At

Tisca

39) Write electron configuration of halogens

Element	Atomic Number	Electronic Configuration
Fluorine	9	$[He]\ 2s^2 2p^5$
Chlorine	17	$[Ne]\ 3s^2 3p^5$
Bromine	35	$[Ar]\ 3d^{10}\ 4s^2\ 4p^5$
Iodine	53	$[Kr]\ 4d^{10}\ 5s^2\ 5p^5$
Astatine	85	$[Xe]\ 4f^{14}\ 5d^{10}\ 6s^2\ 6p^5$

The general valence shell of group 17 is ns^2np^5 only 1 electron is required to get octet electron configuration.

40) Define noble gases

- **Group 18 or VIIIA** elements of the periodic table are called **noble gases.**

41) What is the position of noble gases in the periodic table?

- Noble gases are found the last group which called **group 18 or VIIIA.**

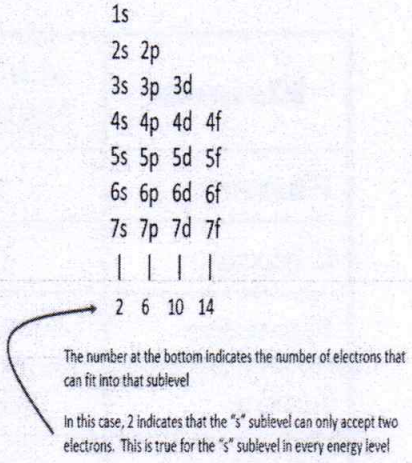
- Write at least physical properties of noble gases
  - Noble gases exist as **gaseous state** in the atmosphere.
  - Noble gases are **non-metals**
  - They are **mono-atomic** in nature
  - Noble gases are **slightly soluble in water**. Their solubility increases down the group.

42) State three common uses of noble gases

- **Krypton** and **xenon** are used in fluorescent bulbs and laser beams.
- **Radon** is used in radiotherapy for treatment of cancer
- **Argon** is used in filament bulbs because the metal filament will no burn in argon.
- **Neon emits** light when high voltage electricity passed through. It is used in **advertising sign**s and fluorine lights.

43) Write electron configuration of noble gases

Element	Atomic Number	Electronic Configuration
Helium	2	$1s^2$
Neon	10	$1s^2 2s^2 2p^6$
Argon	18	$1s^2 2s^2 2p^6 3s^2 3p^6$
Krypton	36	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6$
Xenon	54	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^6$
Radon	86	$1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 4f^{14} 5s^2 5p^6 5d^{10} 6s^2 6p^6$



Note: noble gases are chemically **stable** because they have **8 valence electrons** which is octet except helium. All have **n2p6** configuration form.

Tisca

Choose and circle the correct answer

1. **Group 13 on the periodic table is**
   - a) **Boron group**
   - b) Noble gasses
   - c) Transition metals
   - d) Inner transition metals
2. **Group 13 on the periodic table has how many valence electrons?**
   - a) **3**
   - b) 8
   - c) 18
   - d) 13
3. **What is the valence electron configuration of group 13**
   - a) ns^2p^2
   - b) **ns^2np^1**
   - c) ns^2p^4
   - d) ns^2p^5

Tisca

4. **Which one is group 13 element?**
   - a) Sn
   - b) **In**
   - c) Cd
   - d) Ge
5. **Which element is metalloid in group 13 element?**
   - a) **B**
   - b) Gn
   - c) Al
   - d) Ti
6. **Group 18 of the periodic table are calld_____**
   - a) Alkali metals
   - b) Alkaline earth metals
   - c) Transition metals
   - d) **Noble gases**

7. **Name the group that contains inert (non-reactive) elements**
   a) **Noble gases**
   b) Halogens
   c) Alkali metals
   d) Alkaline earth metals
8. **Group 17 on the periodic table are called**
   a) Transition metals
   b) Metalloids
   c) **Halogens**
   d) Alkali metals
9. Which group has the most reactive non-metal?
   a) Alkali metals
   b) Noble gases
   c) Metalloids
   d) **Halogens**

Tisca

10. **Some groups of the periodic table are more likely to contain elements that are gases than other groups. Which of the following groups contains the greatest number of gaseous elements?**
    a) IIA Group 2
    b) IVA Group 4
    c) VIA Group 6
    d) **VIIIA group 8**
11. **Which of the following is not true for halogens?**
    a) They are non-metals
    b) They show the -1 oxidation number in most in their compounds
    c) **The electronic configuration of their outermost electrons is $ns^2\ np^6$**
    d) Their compounds with metals are generally ionic in nature
    e) Elemental halogens exist as diatomic molecules

CHEMISTRY FORM FOUR

QUESTION AND ANSWERS

CHAPTER SIX

CHEMICAL KINETICS

1) **Define chemical kinetic**
   - The branch of chemistry that deals with the study of the rate of the reactions, factors affecting chemical rates and mechanisms by which the reaction occurs is called **chemical kinetics.**
2) **Why do we study chemical kinetics?**
   - Because it explains the rate of chemical reactions such as: **how quickly medicine** is able to work and how rapidly food spoils.
3) **What is a chemical rate**?
   - The speed of the reaction at which the reactants disappear or products appear is called **reaction rate.**
4) **List factors affecting reaction rates**

The main factors that affect reaction rates are

Tisca

- Nature of reactants
- Temperature
- Concentration of reactants
- Surface area
- The presence of catalyst

5) **Explain how concentration, surface area, temperature, and the addition of catalysts affect reaction rate**
   - When the **concentrations of the reactants** are raised, the reaction proceeds more quickly. This is due to an increase in the number of molecules that have the minimum required energy.
   - When solids and liquids react, increasing the **surface area** of the solid will increase the reaction rate. A decrease in particle size causes an increase in the solid's total surface area.

- Raising the **reaction temperature** by 10 °C can double or triple the reaction rate. This is due to an increase in the number of particles that have the minimum energy required. The reaction rate decreases with a decrease in temperature.
- **Catalysts** can lower the activation energy and increase the reaction rate without being consumed in the reaction.

6) **What is the difference between Catalyst and activation energy?**
   - **Catalyst** is a substance that increases the rate of a chemical reaction without being consumed in the process.
   - **activation energy** is the minimum amount of energy that molecules must have in order for a reaction to occur upon collision.

7) **What is the significance if the surface area is increased?** Tisca
   - More particles are exposed to other reactants
   - There will be more collisions
   - The rate of reaction increases

8) **What is the product formed when hydrogen peroxide decomposes H_2O_2?**
   - **Hydrogen peroxide** decomposes into water and oxygen gas in the form of **foam.**

Hydrogen Peroxide

$$2H_2O_2 \longrightarrow 2H_2O + O_2$$

9) **Define rate law**
   - **Rate law** is an experimentally determined equation that expresses the rate of chemical reactions in terms of molecular concentrations of the reactant.

10) **What is rate constant (K)?**
   - **Rate constant (K)** is the specific rate of reaction when the molar concentrations of the reactants are taken to be unity.

11) **Define order of a reaction**
   - **Order of a reaction** is the experimental exponent of a reactant in the rate equation.

12) What are the conditions required for reaction to occur?

- **Collision:** the reactants must collide
- **Orientation**: the reactants must align properly to break down and form bonds
- **Energy:** the collision must give the energy of activation.

13) What does collision theory provide us?

- **Collision theory** provides an explanation of how particles interact to cause a reaction and the formation of new products.

14) What does collision theory states?

- **Collision theory** states that the peed of chemical reaction is proportional to the number of collisions between reactants.

Tisca

15) What is mechanism of reaction?

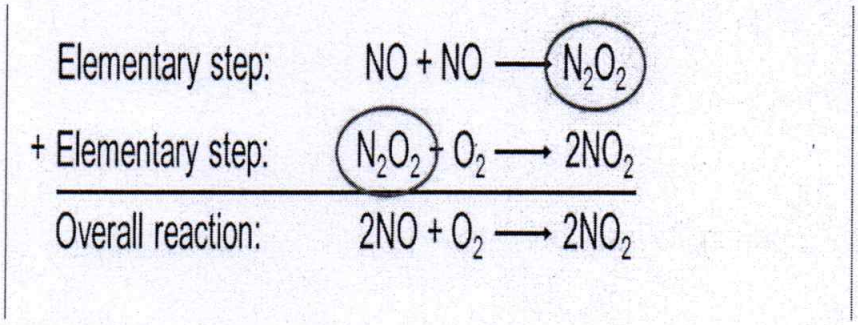
- **Mechanism reaction** is the sequence of elementary steps that together make a complete chemical reaction.

16) What is elementary step in reaction?

- *An **elementary step*** (or elementary reaction) is one step in a series of simple reactions that show the progress of a reaction at the molecular level.

17) What is an intermediate step?

- An **intermediate** *is a species which appears in the mechanism of a reaction, but not in the overall balanced equation*



CHEMISTRY FORM FOUR

QUESTION AND ANSWERS

CHAPTER SEVEN

CHEMICAL EQUILIBRIUM

1) **Define irreversible reaction and reversible reactions**
   - **Irreversible reaction** is reaction that can occur only in one direction.
   - **Reversible reaction** is a reaction that can proceed in both forward and backward directions.
2) **What is chemical equilibrium?**
   - Chemical equilibrium is a state in which the rates of forward reactions is equal to the rate of reverse reactions at constant concentration of reactants and products.
3) **What are the possibilities of the reaction?**

There are three possible ways a reaction can go:

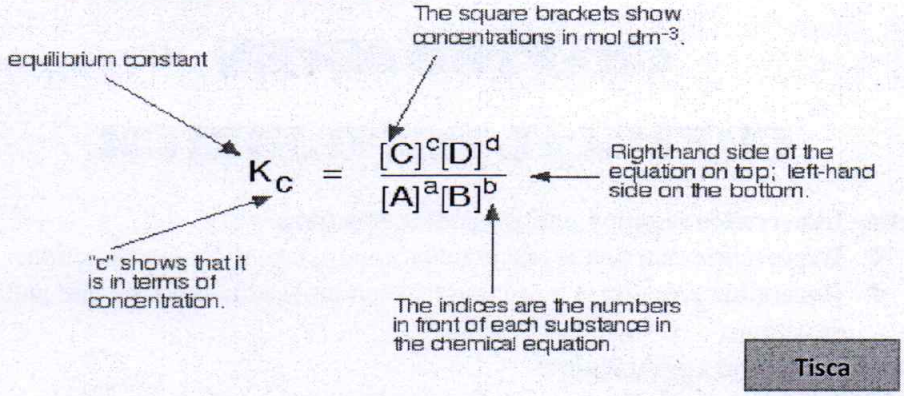
- Reaction goes to completion
- No reaction takes place
- Reaction that appears to stop before reaching completion

4) **What are conditions of chemical equilibrium?**
   - The reaction must occur in closed system
   - The reaction must be reversible
   - The rate of forward must be equal to the rate of the reverse reaction

Tisca

5) **On the bases of phase. List types of chemical equilibrium**
   - Homogeneous chemical equilibrium
   - Heterogeneous chemical equilibrium
6) **Define homogeneous chemical equilibrium**
   - **Homogeneous equilibrium** is a type of equilibrium that the reactants and products are the same phase.
7) **What is heterogamous equilibrium?**
   - **Heterogeneous equilibrium** is a type of equilibrium that the reactants and products are the different phase.

8) **Define equilibrium constant (K_c)**
   - **The equilibrium constant (K_c)** is the ratio of the equilibrium concentrations of the products over the equilibrium concentration of reactants raised to the power of their stoichiometric coefficients.



This expression is known as **equilibrium constant expression.**

Note: if the value of **Kc** is **greater than 1, products are favored**

- ✓ If the value of **Kc** is **less than 1, reactants are favored**
- ✓ if the value of **Kc is equal** 1. The reaction is at **equilibrium** neither **reactants nor products are favored**

9) **Define reaction quotient (Q)**
   - **Reaction quotient** is a quantity that provides a measurement of the relative amounts of products and reactants present in a reaction at a given time.

Note: If **Q** is greater **K** the concentration of **products** is too high

- ✓ If **Q** is less than **K** the concentration of **product**s is too low
- ✓ If **Q** is equal K the system is at **equilibrium**

10) **What does Le Châtelier's principle states?**

- **Le Châtelier's principle states that:** when a system at equilibrium is stressed the system works to restore equilibrium.

11) **What are the effects of concentration on the equilibrium system according to the Le Châtelier's principle?**

- ✓ **Increasing the concentration of a reactant** causes the equilibrium to **shift to the right** producing more products.
- ✓ **Increasing the concentration of a product** causes the equilibrium to **shift to the left** producing more reactants.
- ✓ **Decreasing the concentration of a reactant** causes the equilibrium to **shift to the left** producing less products.
- ✓ **Decreasing the concentration of a product** causes the equilibrium **to shift to the right** producing more products.

12) **What are the effects of changes of pressure on a system at equilibrium?**

- **Increasing pressure (decreasing volume) shifts the equilibrium to the smaller number of moles.**
- **Decreasing pressure (increasing volume) shifts the equilibrium towards the larger number of moles.**

13) **What are the effects of changes of temperature on a system at equilibrium?**

According to lechatlier's principle, Tisca

- when the temperature increases in the case of **exothermic reactions**, the equilibrium shifts towards the reactant side **(Left side),** the products and equilibrium constant decreases.
- For **endothermic reaction** the equilibrium shifts to the product side **(right side)** the product and equilibrium constant increase with increase in temperature.
  - **Exothermic reaction** is reaction that releases energy
  - **Endothermic reaction is** a reaction that absorbs energy.

14) **What is a catalyst?**

A catalyst is a substance that increases the rate of a chemical reaction but is, itself, left unchanged, at the end of the reaction?

Solve the following problems

1) **Calculate for the reaction rate $A_{(g)} + B_{(g)} \longrightarrow C_{(g)} + D_{(g)}$ given that the initial concentrations [A] = 0.10M and [B] = 0.20M and given the reaction constant**

k = 0.015 L/ mol .S

Solution

[A]=0.10M

[B] 0.20M

R = k [A][B]

R = 0.015mol/l.s× 0.10mol/ ×0.20mol/l

R = 0.0003mol/l.s

2) **Write the equilibrium constant expression of the following equation**

$$2NO_2(g) \leftrightarrow 2NO(g) + O_2(g)$$

Solution

$$K = \frac{[NO][O2]}{[NO2]^2}$$

3) **Write the equilibrium constant expression of the following equation**

$CO_{(g)} + 3H2_{(g)} \rightleftarrows CH_4(g) + H_2O(g)$

Solution

$$K = \frac{[CH_4][H_2O]}{[CO][H2]^3}$$

4) write the equilibrium constant expression of the following equation

$2TiCl3(s) + 2HCl\ (g) \rightleftarrows 2TiCl_{4(s)} + H_2\ (g)$

Solution

$$K = \frac{[H_2]}{[HCl]^2}$$

Note: the **two solids** are **omitted** and the coefficients become exponents.

5) write the equilibrium constant expression of the following equation

$P_{4(s)} + 6\ Cl_2\ (g) \rightleftarrows 4\ PCl_{3(s)}$

Solution

$$K = \frac{1}{[Cl_2]^6}$$

6) Determine the value of K when the equilibrium concentrations are: $[SO_2] = 1.20M$; $[NO_2] = 0.60M$; $[NO] = 1.6\ M$; $[SO_3] = 2.2\ M$...

$$SO_2\ (g) + NO_2\ (g) \rightleftarrows SO_3\ (g) + NO\ (g)$$

Tisca

Solutions

$[SO_3] = 2.2\ M$

$[NO] = 1.6\ M$

$[SO_2] = 1.20M$

$[NO_2] = 0.60M$

$$K = \frac{[SO_3][NO]}{[SO_2][NO_2]}$$

$$K = \frac{(2.2M)(1.6M)}{(1.20M)(0.60M)}$$

$$\boxed{K = 4.9}$$

7) **Calculate concentration of CO if [H_2O] = 0.100 M; [H_2] = 0.100 M; [CO_2] = 0.100 M. and**

$$CO_{(g)} + H_2O_{(g)} \rightleftharpoons H_{2(g)} + CO_{2(g)} \qquad K = 1.34$$

Solutions

[CO_2] = 0.100 M
[H_2] =0.100M
[CO] = ?
[H_2O] = 0.100M
K=1.34

$$K = \frac{[H_2][CO_2]}{[CO][H_2O]}$$

$$= [CO] = \frac{[H_2][CO_2]}{K[H_2O]}$$

$$[CO] = \frac{(0.100M)(0{,}100M)}{(1.34)(0.100M)}$$

$$\boxed{[Co] = 0.746M}$$

Choose the correct answer

1) **System of reaction in which no reactant leaves reaction mixture, is termed as:**
   a) Open system
   b) **Closed system**
   c) Semi-open system
   d) Partially closed system
2) **Reactions in which products cannot turn back into reactants are termed as**
   a) **Irreversible reactions**
   b) Reversible reactions
   c) Equilibrium reactions
   d) Redox reactions
3) **On increasing concentration of one or more reactants, more products will be generated until**
   a) **Equilibrium**
   b) Closed system
   c) Open system
   d) All of Above
4) **Position of equilibrium explains amounts of**
   a) Products
   b) Reactants
   c) Catalyst
   d) **Both A and B**
5) **Reversible reactions are those which products react together to give original**
   a) **Reactants**
   b) Products
   c) Enzymes
   d) All of Above

Tisca

6) In equilibrium state, backward and forward reactions take place at
   a) **Same time**
   b) Simultaneously
   c) One time
   d) All of Above

7) Le chateler's principle states that if one or more factors disturbs equilibrium, position of equilibrium will shift where change will be
   a) **Reduced**
   b) Induced
   c) Increased
   d) Oxidized

8) System will not be in equilibrium when there is an increase in concentration of
   a) **Reactants**
   b) Products
   c) Catalysts
   d) Activation energy

9) Catalysts speed up reactions but when equilibrium is reached it has
   a) Huge effect
   b) **No effect**
   c) Slow effect
   d) Ended

CHEMISTRY FORM FOUR
QUESTIONS AND ANSWERS
CHAPTER EIGHT
NUCLEAR CHEMISTRY

1) **What is a nuclear chemistry?**
   - **Nuclear chemistry** is the study of the structure of **atomic nuclei** and the **changes** they undergo.
   - **Nuclear chemistry** is the reaction which involves changes in atomic nuclei.
2) **Why the study of nuclear chemistry is so important?**
   - Nuclear changes are sources of **radioactivity** and **nuclear power.** That is why nuclear chemistry is very important branch of chemistry
3) **What is radioactivity?**
   - **Radioactivity** is a phenomenon of spontaneous and uncontrollable disintegration

Occurs by emission of active radiations from unstable atomic nucleus.

4) **Define radioisotopes**
   - Atoms which emit radioactive radiations are called **radioisotopes**
5) **What are types of nuclear radiations**
   - There are three types of nuclear radiations namely

Tisca

   a) **Alpha rays (α)**
   b) **Beta rays (β)**
   c) **Gamma rays (γ)**

6) **What is atom?**
   - **Atom** is the smallest particle of matter that can take part in a chemical reaction.
7) **List the three sub-atomic particles.**
   - The three sub-atomic particles of an atom are: **Protons, electrons** and **neutrons**.
8) **What is atomic number?**
   - **Atomic number:** - is the number of protons in the nucleus of an atom.

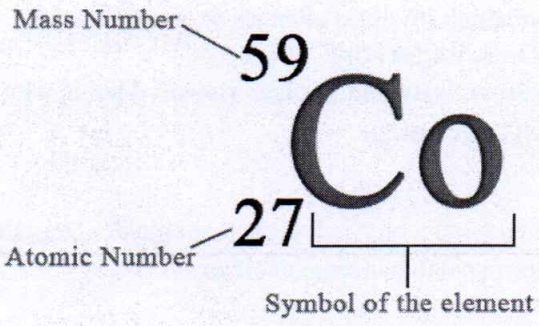
9) What is a mass number?

- **Mass number: -** is the sum of the number of protons and neutrons in the nucleus of an atom. It can also be called **nucleon number.**

[A = Z + N]

10) Where does the atomic number and mass number can be indicated in the symbol of an element?

- The mass number, **A** is placed as left superscript of the symbol, while the atomic number, **Z** is placed as a left subscript of the symbol. As shown below:



Tisca

11)

12) Define Isotope

- **Isotopes** are atoms of the same element that have same atomic number but different mass number or neutron number. E.g. Hydrogen has 3 **isotopes**
- 1) ${}_{1}^{1}H$ **(protium)** 2) ${}_{1}^{2}H$ **(deuterium)** 3) ${}_{1}^{3}H$ **(tritium)**
  - Carbon has three **isotopes:** ${}_{6}^{12}C$, ${}_{6}^{13}C$, ${}_{6}^{14}C$

13) What caused by nuclear reactions?

- **Nuclear reaction** can be brought about by the interaction of two nuclei or under the impact of sub-atomic particle on the nucleus.

*14) W*hat is the difference between nuclear reaction and chemical reaction?

Nuclear reaction	Chemical reactiom
1. Occurs when nucliec combine, split and emit radiaton 2. Can involve protons , neutroons and electrons 3. Releases large amount of heat energy. 4. Not effected by environmental factors such as temperature.	1. Ocuurs when bonds are broken and formed 2. Involve only valence electrons 3. Associates with small energy changes 4. Are affected by environmental factors such as temperature and pressure.

Tisca

15) Where is generated from the force that holds the nucleus together?

❖ The force which holds the nucleus together is generated by nuclear binding energy.

16) What is a mass defect?

❖ The difference between the mass of the atom and the sum of the masses of its parts is called the ***mass defect*** **(Δm).**

17) What is due to the mass defect?

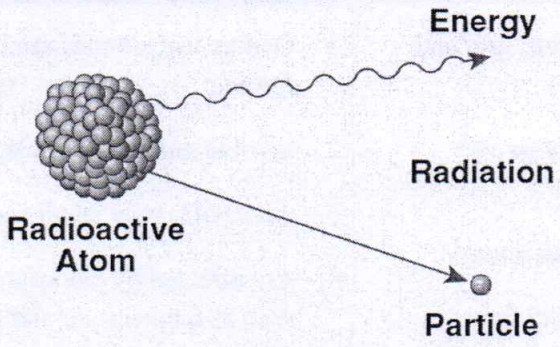
❖ The loss in mass or mass defect is due to the conversion of mass to biding energy when the nucleus is formed.

18) What is binding energy?

❖ **Nuclear binding energy** is the energy required to separate or combine an atomic

19) What is a radioactive decay?

- **Radioactive decay**: is the process by which unstable atoms release particles and energy in an effort to become stable.



20) List the types of radioactive decay.

- **Alpha decay, Beta decay and gamma decay**

21) Define alpha decay?

- Whenever the element emits **α-particle** the **atomic mass decrease by 4** units and the **atomic number decreases by 2 units**. This is known as alpha decay**. Examples:**

a) $^{234}_{92}U \xrightarrow{\alpha-decay} {}^{230}_{90}Th + {}^{4}_{2}He$

b) $^{230}_{90}Th \xrightarrow{\alpha-decay} {}^{226}_{88}Ra + {}^{4}_{2}He$

c) $^{226}_{88}Ra \xrightarrow{\alpha-decay} {}^{222}_{86}Rn + {}^{4}_{2}He$

d) $^{222}_{86}Rn \xrightarrow{\alpha-decay} {}^{218}_{84}Po + {}^{4}_{2}He$

e) $^{218}_{84}Po \xrightarrow{\alpha-decay} {}^{214}_{82}Pb + {}^{4}_{2}He$

Tisca

Note: α -decay involves emission of helium nucleus.

22) What is beta decay?

- Whenever the element emits a **β-particle** the **mass number remains unchanged** and the atomic number **increased by 1 unit.** This is known as beta particle.
- **Examples:**

1) $^{234}_{90}Th \xrightarrow{\beta-dacay} ^{234}_{91}Pa + {}^{0}_{-1}\beta$

2) $^{210}_{81}Tl \xrightarrow{\beta-decay} ^{210}_{82}Pb + {}^{0}_{-1}\beta$

3) $^{210}_{83}Bi \xrightarrow{\beta-decay} ^{210}_{84}Po + {}^{0}_{-1}\beta$

4) $^{214}_{82}Pb \xrightarrow{\beta-decay} ^{214}_{83}Bi + {}^{0}_{-1}\beta$

23) What is a gamma decay?

- Whenever the element emits a **γ-rays** there is no change either in **mass number** or **atomic number.** This is known as **γ-rays. Examples:**

1) $^{81}_{36}Kr \xrightarrow{\gamma-decay} ^{81}_{36}Kr + {}^{0}_{0}\gamma$

2) $^{60}_{27}Co \xrightarrow{\gamma-decay} ^{60}_{27}Co + {}^{0}_{0}\gamma$

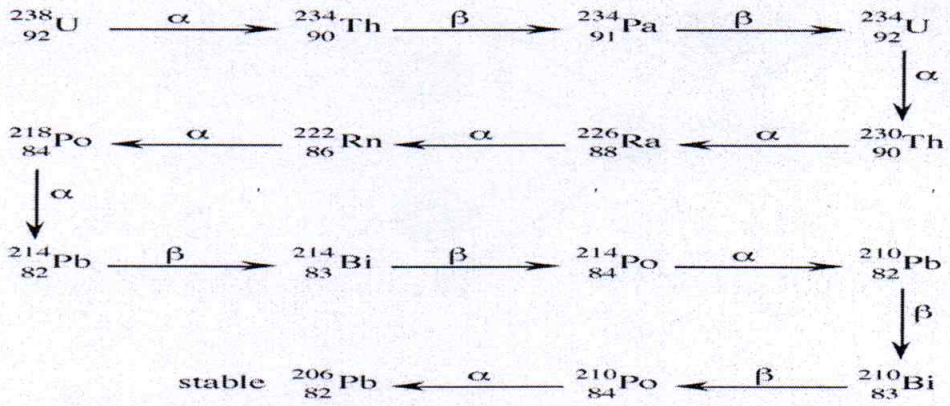
Tisca

Note: γ -decay involve radiation of higher energy photon.

24) Define decay series with example.

- Often, a radioactive nucleus cannot reach a stable state through a single decay. In such cases, a series of decays will occur until a stable nucleus is formed.
- The decay of U-238 is an example of this.

The U-238 decay series starts with U-238 and goes through fourteen separate decays to finally reach a stable nucleus, Pb-206.



25) What are types of nuclear reactions?

There are four types of nuclear reactions

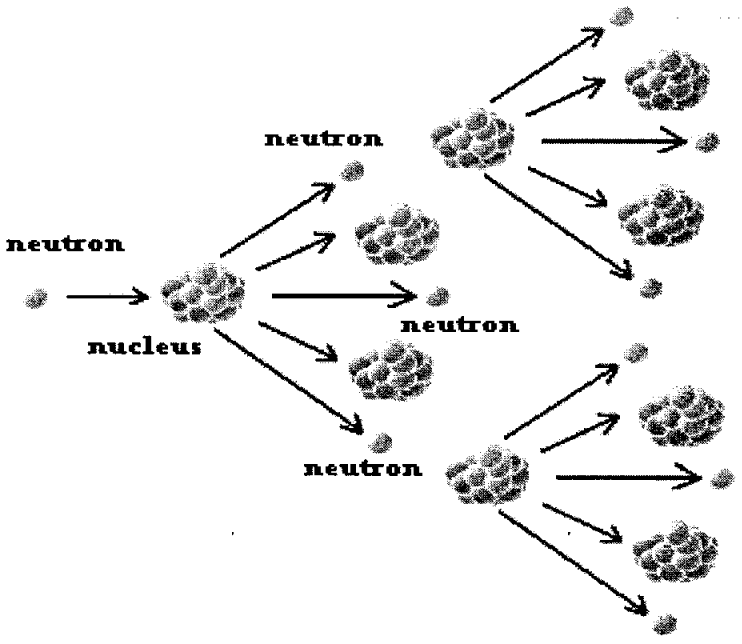
- Nuclear fission
- Nuclear fusion
- Radioactive or nuclear decay
- Transmutation

26) What is the difference between nuclear fission and nuclear fusion?

- **Nuclear Fission** = is the splitting of nuclei
- **Nuclear Fusion** = **is** the joining of nuclei (they fuse together)

27) What is a chain reaction?

- **Chain reaction:** is the continuing series of reactions in which each produces a product that can react again.



28) **What is transmutation?**

- **Transmutation** is a change in the identity of a nucleus as a result of a change in the number of protons.

29) **Define half-life?**

- **Half-life** is the time required for half of the original sample of nuclei to decay
- **Half-life** is the time it takes for half (50%) of the radioactive nuclei to decay to the daughter nuclide.

30) **The half-life of radium-226 is about 1600years.**
How many grams of a 2.00gram sample will remain after 4800 years?

Solution

$T_{\frac{1}{2}} = 1600yrs$

$M_O = 2g$

$T = 4800yrs$

$Mt = ?$

$$Mt = N_O\left(\frac{1}{2}\right)^N$$

$$Mt = (2g)\left(\frac{1}{2}\right)^{\frac{4800\,yrs}{1600\,yrs}}$$

$$Mt = (2g)\left[\left(\frac{1}{2}\right)\right]^3$$

$$Mt = (2g)\left(\frac{1}{8}\right)$$

$$Mt = \frac{2}{8} = 0.25g$$

31) **Define Rem**

- The unit most commonly used to express radiation is called **rem**.

32) **List types of radiation**

- Radiation can be either ionizing radiation and non-ionizing radiation

33) **What occurs when nuclear fission occurs?**

- Nuclear fission causes three events
- The release of radiation
- Release of neutrons (usually two or three)
- Formation of two new nuclei (fission products)

Tisca

34) List application of nuclear chemistry

Nuclear chemistry is used

- **In medicine:** radioactive sources are used to diagnose, treat disease and to sterilize medical instruments.
- **In Agriculture:** Radioactive materials are used to improve food crops, preserve food, measure soil moisture content, erosion rate and to control insects.
- **In industrial development:** improving productivity, to monitor all steps of industrial process and safety control measure.
- **In energy:** produce energy for heat, electrical power and nuclear weapons.
- **In archeology:** we use to determine the age of rocks or archeological relics.
- **In environment:** we use to detect smoke and to monitor pollutants.

Tisca